MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3320 LOW GRADE THICKENING
DESIGN OF STAIR FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The purpose of this document is to analysis and design of **LOW GRADE THICKENING STAIR** foundation. In this document design & analysis of **STAIR** foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Load data for foundation design taken from document no. **MD-216-3320-EG-ST-CAL-2185.**

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

LOAD 1 = DEAD LOAD (DL)

LOAD 2 = LIVE LOAD (LL)

LOAD 3 = WIND LOAD + X (WLX1)

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LOAD 4 = WIND LOAD - X (WLX2)

LOAD 5 = WIND LOAD + Z (WLZ1)

LOAD 6 = WIND LOAD - Z (WLZ2)

LOAD 7 = SOIL LOAD (HL)

4.3 Load combination

A. The following load combination are considered for member strength calculation.

101 1.4DL

102 1.4DL+1.7LL

103 1.2DL+1.0LL+1.6WLX1

104 1.2DL+1.0LL+1.6WLX2

105 1.2DL+1.0LL+1.6WLZ1

106 1.2DL+1.0LL+1.6WLZ2

107 0.9DL+1.6HL+1.6WLX1

108 0.9DL+1.6HL+1.6WLX2

109 0.9DL+1.6HL+1.6WLZ1

110 0.9DL+1.6HL+1.6WLZ2

B) The following load combination are considered for base pressure, sliding & overturning check.

201 1.0DL+1.0HL+1.0LL

202 1.0DL+1.0HL+0.75LL

203 1.0DL+1.0HL+1.0WLX1

204 1.0DL+1.0HL+1.0WLX2

205 1.0DL+1.0HL+1.0WLZ1

206 1.0DL+1.0HL+1.0WLZ2

207 1.0DL+1.0HL+0.75LL+0.75WLX1

208 1.0DL+1.0HL+0.75LL+0.75WLX2

209 1.0DL+1.0HL+0.75LL+0.75WLZ1

210 1.0DL+1.0HL+0.75LL+0.75WLZ2

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211 0.6DL+1.0HL+1.0WLX1

212 0.6DL+1.0HL+1.0WLX2

213 0.6DL+1.0HL+1.0WLZ1

214 0.6DL+1.0HL+1.0WLX1

4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1.5m depth below EGL (RL 912.000) refer Table 5 - BH-26 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 60000 kN/m^2/m at 1.5m depth below EGL (RL 912.000) refer Table 5 - BH-26 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5.1 DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-Z	Along-X
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	6.00m	1.50m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.30	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

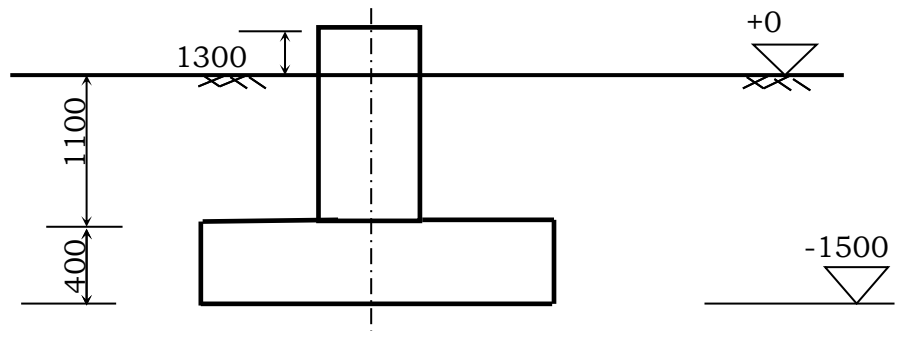
Calculation of Weight of Foundation :

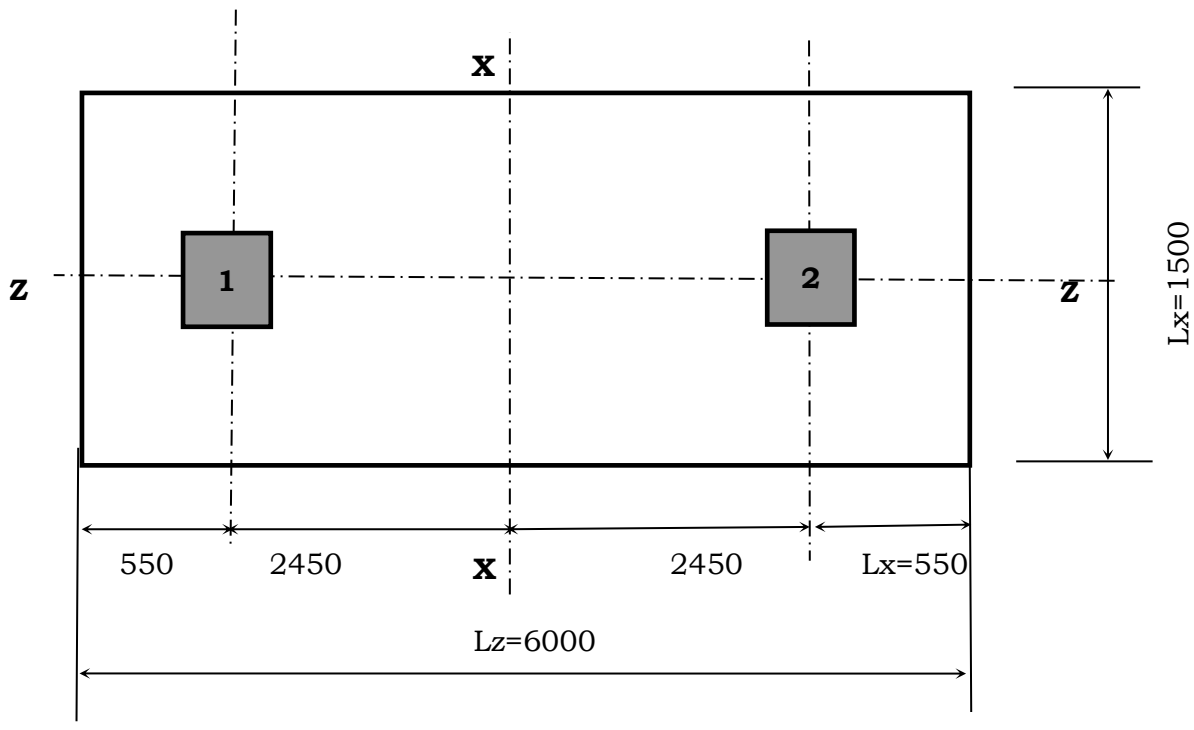
Weight of raft = (6*1.5*0.4*25) = 90.00KN

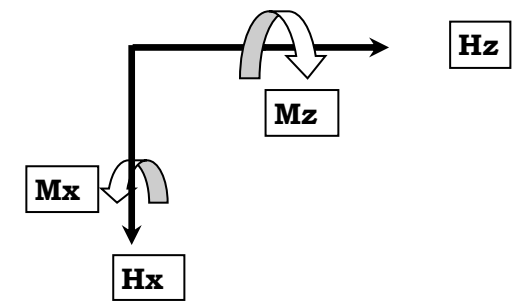
Weight of soil = ((6*1.5)-2*(0.5*0.5))*1.1*18 = 168.30KN

Total = 258.30KN

Submerged wt. = 258.3-6*1.5*1.5*10 = 123.30KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0LL	60.0	0.0	0.0	10.0	28.0	110.0	0.0	0.0	-10.0	-28.0
202 1.0DL+0.75LL	50.0	0.0	0.0	7.5	21.0	90.0	0.0	0.0	-7.5	-21.0
203 1.0DL+1.0WLX1	20.0	0.0	0.0	0.0	0.0	30.0	0.0	0.0	0.0	0.0
204 1.0DL+1.0WLX2	20.0	0.0	0.0	0.0	0.0	30.0	0.0	0.0	0.0	0.0
205 1.0DL+1.0WLZ1	-10.0	0.0	0.0	-20.0	-56.0	60.0	0.0	0.0	-20.0	-56.0
206 1.0DL+1.0WLZ2	50.0	0.0	0.0	20.0	56.0	0.0	0.0	0.0	20.0	56.0
207 1.0DL+0.75LL+0.75WLX1	50.0	0.0	0.0	7.5	21.0	90.0	0.0	0.0	-7.5	-21.0
208 1.0DL+0.75LL+0.75WLX2	50.0	0.0	0.0	7.5	21.0	90.0	0.0	0.0	-7.5	-21.0
209 1.0DL+0.75LL+0.75WLZ1	27.5	0.0	0.0	-7.5	-21.0	112.5	0.0	0.0	-22.5	-63.0
210 1.0DL+0.75LL+0.75WLZ2	72.5	0.0	0.0	22.5	63.0	67.5	0.0	0.0	7.5	21.0
211 0.6DL+1.0WLX1	12.0	0.0	0.0	0.0	0.0	18.0	0.0	0.0	0.0	0.0
212 0.6DL+1.0WLX2	12.0	0.0	0.0	0.0	0.0	18.0	0.0	0.0	0.0	0.0
213 0.6DL+1.0WLZ1	-18.0	0.0	0.0	-20.0	-56.0	48.0	0.0	0.0	-20.0	-56.0
214 0.6DL+1.0WLZ2	42.0	0.0	0.0	20.0	56.0	-12.0	0.0	0.0	20.0	56.0

Check for overturning and sliding:

	Resultant Load at bottom of footing								
						Overturning		Sliding	
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0LL	428.3	0.00	122.5	0.0	0.0	NA	10.5	NA	NA
202 1.0DL+0.75LL	398.3	0.00	98.0	0.0	0.0	NA	12.2	NA	NA
203 1.0DL+1.0WLX1	308.3	0.00	24.5	0.0	0.0	NA	37.8	NA	NA
204 1.0DL+1.0WLX2	308.3	0.00	24.5	0.0	0.0	NA	37.8	NA	NA
205 1.0DL+1.0WLZ1	308.3	-40.00	59.5	0.0	0.0	NA	15.5	NA	3.1
206 1.0DL+1.0WLZ2	308.3	40.00	-10.5	0.0	0.0	NA	88.1	NA	3.1
207 1.0DL+0.75LL+0.75WLX1	398.3	0.00	98.0	0.0	0.0	NA	12.2	NA	NA
208 1.0DL+0.75LL+0.75WLX2	398.3	0.00	98.0	0.0	0.0	NA	12.2	NA	NA
209 1.0DL+0.75LL+0.75WLZ1	398.3	-30.00	124.3	0.0	0.0	NA	9.6	NA	5.3
210 1.0DL+0.75LL+0.75WLZ2	398.3	30.00	71.8	0.0	0.0	NA	16.7	NA	5.3
211 0.6DL+1.0WLX1	288.3	0.00	14.7	0.0	0.0	NA	58.8	NA	NA
212 0.6DL+1.0WLX2	288.3	0.00	14.7	0.0	0.0	NA	58.8	NA	NA
213 0.6DL+1.0WLZ1	288.3	-40.00	49.7	0.0	0.0	NA	17.4	NA	2.9
214 0.6DL+1.0WLZ2	288.3	40.00	-20.3	0.0	0.0	NA	42.6	NA	2.9
					Min.=	0.0	9.6	0.0	2.9

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	81.05	0
Min Px	1	201	0	81.05	0
Max Py	2	209	0	126.89	0
Min Py	1	213	0	0.00	0
Max Pz	1	201	0	81.05	0
Min Pz	1	201	0	81.05	0

Gross Maximum Base pressure, p_{max} = 126.9 KN/m^2

Net Allowable Safe Bearing Capacity = 500 KN/m^2 (Bore hole 26)
Depth of soil above footing = 1.1 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 519.8 KN/m^2
> 126.9 KN/m^2
Hence OK

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Design of foundation raft F1

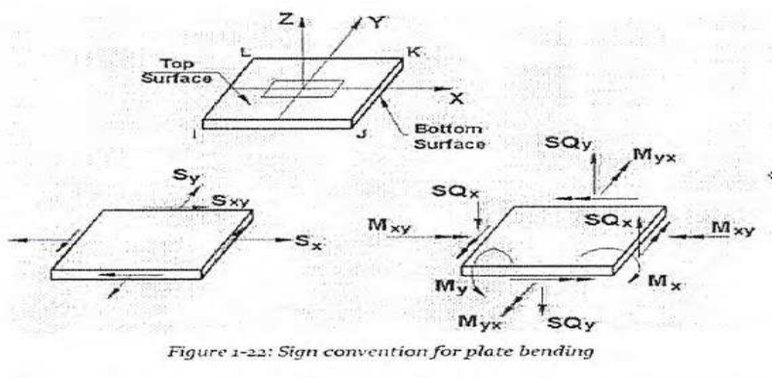
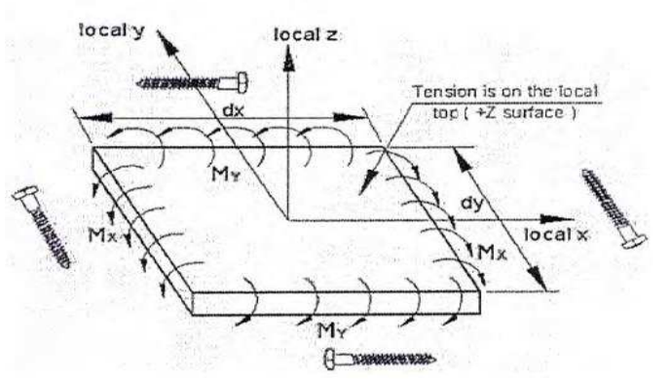


Figure 1-22: Sign convention for plate bending



Sign convention :

Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :

Mx : BM along X direction, +ve means tension on +Z surface of plate

My : BM along Y direction, +ve means tension on +Z surface of plate

For -ve Mx & My it is vice-virsa

Shear force :

SQx : Shear at face perpendicular to X axis

SQy : Shear at face perpendicular to Y axis

Axial force :

+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	64	0.283	-0.031	0	0	0	-17.38	-2.317	4.173	-21.556	-6.49	105 1.2DL+1.0LL+1.6WLZ1
Min Qx	53	-0.283	-0.031	0	0	0	-17.38	-2.317	0	-17.383	-2.317	105 1.2DL+1.0LL+1.6WLZ1
Max Qy	171	0.065	0.316	0	0	0	-5.027	30.123	7.358	-12.385	37.481	105 1.2DL+1.0LL+1.6WLZ1
Min Qy	247	0.028	-0.303	0	0	0	-3.415	32.048	-6.433	-9.848	38.481	106 1.2DL+1.0LL+1.6WLZ2
Max Sx	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Min Sx	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Max Sy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Min Sy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Max Sxy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Min Sxy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Max Mx	45	-0.088	-0.106	0	0	0	13.85	6.668	2.716	16.566	9.384	109 0.9DL+1.6HL+1.6WLZ1
Min Mx	62	0.14	-0.203	0	0	0	-34.99	-10.435	2.192	-37.186	-12.627	105 1.2DL+1.0LL+1.6WLZ1
Max My	174	-0.08	-0.033	0	0	0	-4.655	82.547	5.75	-10.405	88.297	105 1.2DL+1.0LL+1.6WLZ1
Min My	242	0.077	-0.158	0	0	0	-6.448	-56.191	2.57	-9.018	-58.761	109 0.9DL+1.6HL+1.6WLZ1
Max Mxy	175	-0.112	0.076	0	0	0	0.33	68.714	9.768	10.098	78.482	105 1.2DL+1.0LL+1.6WLZ1
Min Mxy	79	0.112	0.076	0	0	0	0.33	68.714	-9.768	10.098	78.482	105 1.2DL+1.0LL+1.6WLZ1

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PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	66	0.082	0.067	0	0	0	-2.22	2.701	1.203	-3.423	3.904	105 1.2DL+1.0LL+1.6WLZ1
Min Qx	51	-0.082	0.067	0	0	0	-2.22	2.701	0	-2.22	2.701	105 1.2DL+1.0LL+1.6WLZ1
Max Qy	173	0.015	0.15	0	0	0	-1.321	41.577	4.664	-5.985	46.241	105 1.2DL+1.0LL+1.6WLZ1
Min Qy	248	0.005	-0.142	0	0	0	-1.111	42.224	-4.243	-5.354	46.467	106 1.2DL+1.0LL+1.6WLZ2
Max Sx	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Min Sx	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Max Sy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Min Sy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Max Sxy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Min Sxy	248	0.01	-0.016	0	0	0	-0.224	0.232	-0.306	-0.53	0.538	101 1.4DL
Max Mx	238	-0.016	0.046	0	0	0	2.979	51.687	-0.24	3.219	51.927	110 0.9DL+1.6HL+1.6WLZ2
Min Mx	178	0.022	0.128	0	0	0	-8.2	-18.74	0.833	-9.033	-19.573	106 1.2DL+1.0LL+1.6WLZ2
Max My	178	-0.015	-0.01	0	0	0	0.616	76.166	0.859	1.475	77.025	105 1.2DL+1.0LL+1.6WLZ1
Min My	238	0.021	-0.092	0	0	0	-5.837	-43.219	-0.214	-6.051	-43.433	109 0.9DL+1.6HL+1.6WLZ1
Max Mxy	173	0.015	0.15	0	0	0	-1.321	41.577	4.664	-5.985	46.241	105 1.2DL+1.0LL+1.6WLZ1
Min Mxy	73	-0.015	0.15	0	0	0	-1.321	41.577	-4.664	-5.985	46.241	105 1.2DL+1.0LL+1.6WLZ1

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9D	16.57	top face	50	342	16.566	0.19	0.0005	154.3	615.6	16	326.74	200	1005.8	14.18	17.73	0.055	tension control
105 1.2D	-37.19	bottom face	50	342	37.186	0.42	0.0010	347.7		16	326.74	200	1005.8	14.18	17.73	0.055	tension control
105 1.2D	10.10	top face	50	342	10.098	0.12	0.0003	93.9		16	326.74	200	1005.8	14.18	17.73	0.055	tension control
105 1.2D	10.10	top face	50	342	10.098	0.12	0.0003	93.9		16	326.74	200	1005.8	14.18	17.73	0.055	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.082	1	32.8	3.423	0.003	0.342	379.762	340.826	586.757	340.83	1257	> 32.8	ok	255.6	φVc>Vu	not required
	Min Qx	0.082	1	32.8	2.22			410.127			340.83	1257	> 32.8	ok	255.6	φVc>Vu	not required

Reinforcement Along Z:

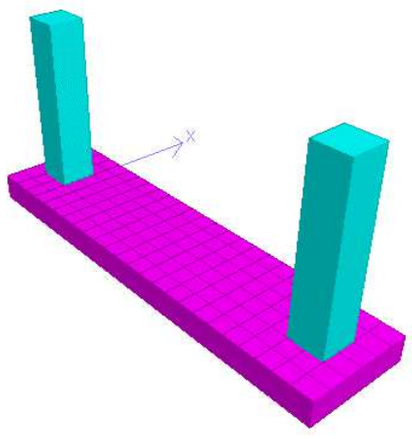
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
105 1.2D	88.30	top face	50	342	88.297	1.01	0.0024	834.0	615.6	16	241.19	200	1005.8	14.18	17.73	0.055	tension control
109 0.9D	-58.76	bottom face	50	342	58.761	0.67	0.0016	551.7		16	326.74	200	1005.8	14.18	17.73	0.055	tension control
105 1.2D	78.48	top face	50	342	78.482	0.89	0.0022	739.8		16	271.89	200	1005.8	14.18	17.73	0.055	tension control
105 1.2D	78.48	top face	50	342	78.482	0.89	0.0022	739.8		16	271.89	200	1005.8	14.18	17.73	0.055	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	105 1.2D	0.15	1	60	46.241	0.003	0.342	331.316	340.826	fc'586.8	331.32	349	> 60	ok	248.5	φVc>Vu	not required
	106 1.2D	0.142	1	56.8	46.467			330.876			330.88	348	> 56.8	ok	248.2	φVc>Vu	not required

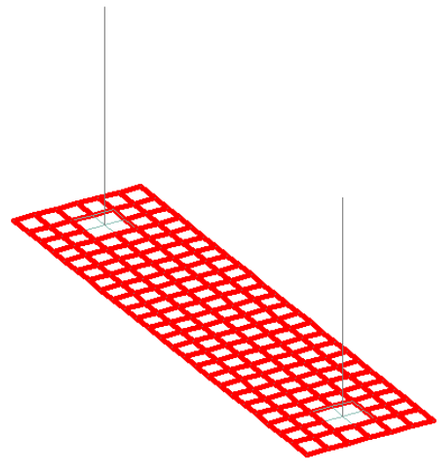
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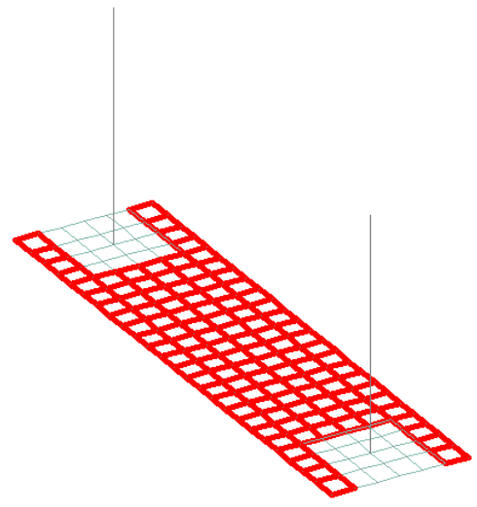
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3D Rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	**200**	c/c
At Top	16ϕT	**200**	c/c
Along Z Direction :			
At Bottom	16ϕT	**200**	c/c
At Top	16ϕT	**200**	c/c

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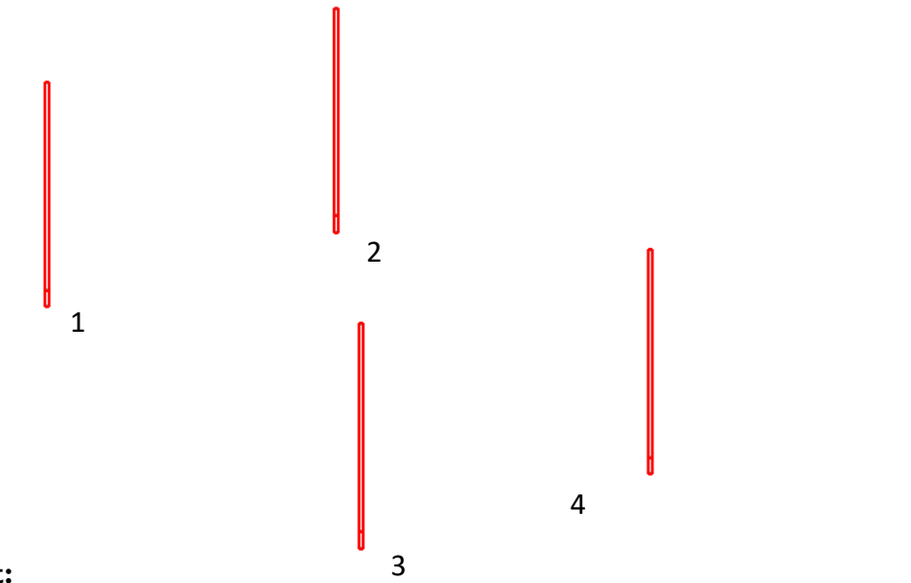
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5.2 DESIGN OF FOUNDATION MKD. F2

(STAAD Node -1 2 3 4)

Size of Columns		Along-X	Along-Z
Node	1	0.50m	0.50m
Node	2	0.50m	0.50m
Node	3	0.50m	0.50m
Node	4	0.50m	0.50m
Size of fdn. raft:		4.00m	6.00m
Thickness of raft:		0.40 m	
Original Gr. Level:		0.00 m	
Finished Gr. Level:		0.00 m	
U/S of B/Pl Above GL		1.30 m	
U/S level of fdn:		-1.50 m	
Unit weight of conc:		25 kN/m^3	
Unit weight of soil:		18 kN/m^3	



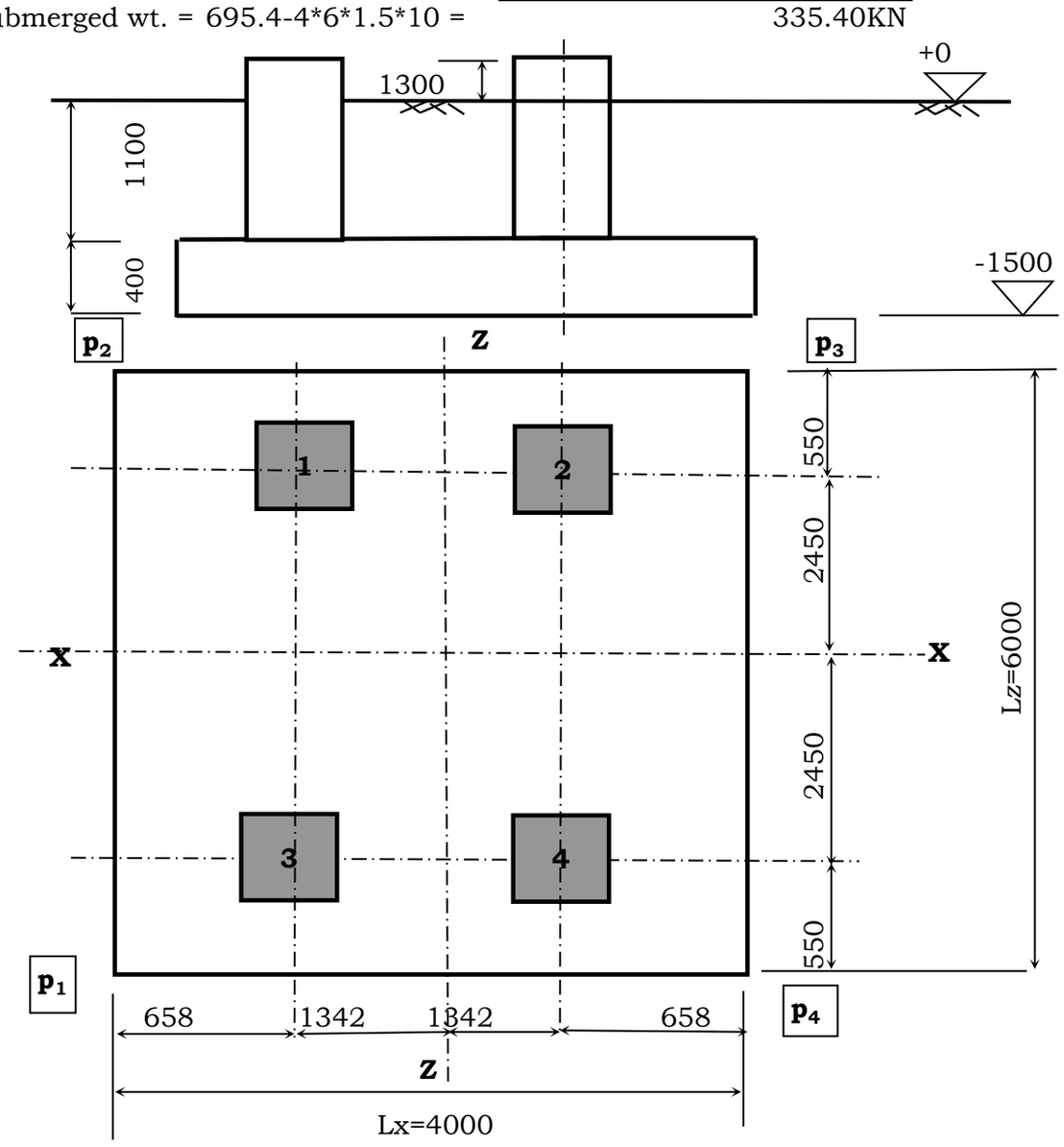
Calculation of Weight of Foundation :

Weight of raft = (4*6*0.4*25) = 240.00KN

Weight of soil = ((4*6)-3*((0.5*0.5)+(0.5*0.5))*1.1*18 = 455.40KN

Total = 695.40KN

Submerged wt. = 695.4-4*6*1.5*10 = 335.40KN

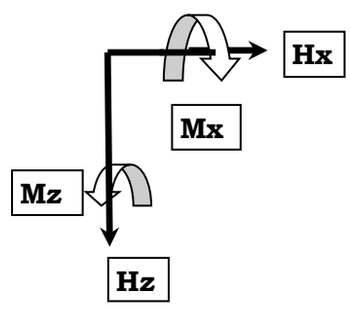


Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 334mm = 0.334m

Effective depth of slab along Z direction = 342mm = 0.342m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F2 :

Node	1					2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0LL	150	0	0	10	28	60	0	0	0	0
202 1.0DL+0.75LL	122.5	0	0	7.5	21	50	0	0	0	0
203 1.0DL+1.0WLX1	-50	-20	56	0	0	110	-20	56	0	0
204 1.0DL+1.0WLX2	130	20	-56	0	0	-70	20	-56	0	0
205 1.0DL+1.0WLZ1	10	0	0	-30	-84	-10	0	0	-20	-56
206 1.0DL+1.0WLZ2	70	0	0	30	84	50	0	0	20	56
207 1.0DL+0.75LL+0.75WLX1	55	-15	42	7.5	21	117.5	-15	42	0	0
208 1.0DL+0.75LL+0.75WLX2	190	15	-42	7.5	21	-17.5	15	-42	0	0
209 1.0DL+0.75LL+0.75WLZ1	100	0	0	-15	-42	27.5	0	0	-15	-42
210 1.0DL+0.75LL+0.75WLZ2	145	0	0	30	84	72.5	0	0	15	42
211 0.6DL+1.0WLX1	-66	-20	56	0	0	102	-20	56	0	0
212 0.6DL+1.0WLX2	114	20	-56	0	0	-78	20	-56	0	0
213 0.6DL+1.0WLZ1	-6	0	0	-30	-84	-18	0	0	-20	-56
214 0.6DL+1.0WLZ2	54	0	0	30	84	42	0	0	20	56

Node	3					4				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0LL	190	0	0	-40	-112	110	0	0	0	0
202 1.0DL+0.75LL	155	0	0	-32.5	-91	90	0	0	0	0
203 1.0DL+1.0WLX1	-60	-30	84	-10	-28	150	-30	84	0	0
204 1.0DL+1.0WLX2	160	30	-84	-10	-28	-90	30	-84	0	0
205 1.0DL+1.0WLZ1	90	0	0	-20	-56	60	0	0	0	0
206 1.0DL+1.0WLZ2	10	0	0	0	0	0	0	0	0	0
207 1.0DL+0.75LL+0.75WLX1	72.5	-22.5	63	-32.5	-91	180	-22.5	63	0	0
208 1.0DL+0.75LL+0.75WLX2	237.5	22.5	-63	-32.5	-91	0	22.5	-63	0	0
209 1.0DL+0.75LL+0.75WLZ1	185	0	0	-40	-112	112.5	0	0	0	0
210 1.0DL+0.75LL+0.75WLZ2	125	0	0	-25	-70	67.5	0	0	0	0
211 0.6DL+1.0WLX1	-80	-30	84	-6	-16.8	138	-30	84	0	0
212 0.6DL+1.0WLX2	140	30	-84	-6	-16.8	-102	30	-84	0	0
213 0.6DL+1.0WLZ1	70	0	0	-16	-44.8	48	0	0	0	0
214 0.6DL+1.0WLZ2	-10	0	0	4	11.2	-12	0	0	0	0

L/C	Load at top of foundation					Load at bottom of foundation				
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0LL	510.0	0.0	-228.1	-30.0	-304.5	1205.4	0.0	-228.1	-30.0	-304.5
202 1.0DL+0.75LL	417.5	0.0	-184.5	-25.0	-247.6	1112.9	0.0	-184.5	-25.0	-247.6
203 1.0DL+1.0WLX1	150.0	-100.0	776.5	-10.0	-101.5	845.4	-100.0	776.5	-10.0	-101.5
204 1.0DL+1.0WLX2	130.0	100.0	-883.9	-10.0	-52.5	825.4	100.0	-883.9	-10.0	-52.5
205 1.0DL+1.0WLZ1	150.0	0.0	-67.1	-70.0	-563.5	845.4	0.0	-67.1	-70.0	-563.5
206 1.0DL+1.0WLZ2	130.0	0.0	-40.3	50.0	409.5	825.4	0.0	-40.3	50.0	409.5
207 1.0DL+0.75LL+0.75WLX1	425.0	-75.0	438.1	-25.0	-266.0	1120.4	-75.0	438.1	-25.0	-266.0
208 1.0DL+0.75LL+0.75WLX2	410.0	75.0	-807.2	-25.0	-229.3	1105.4	75.0	-807.2	-25.0	-229.3
209 1.0DL+0.75LL+0.75WLZ1	425.0	0.0	-194.6	-70.0	-612.5	1120.4	0.0	-194.6	-70.0	-612.5
210 1.0DL+0.75LL+0.75WLZ2	410.0	0.0	-174.5	20.0	117.3	1105.4	0.0	-174.5	20.0	117.3
211 0.6DL+1.0WLX1	94.0	-100.0	798.0	-6.0	-70.7	789.4	-100.0	798.0	-6.0	-70.7
212 0.6DL+1.0WLX2	74.0	100.0	-862.4	-6.0	-21.7	769.4	100.0	-862.4	-6.0	-21.7
213 0.6DL+1.0WLZ1	94.0	0.0	-45.6	-66.0	-532.7	789.4	0.0	-45.6	-66.0	-532.7
214 0.6DL+1.0WLZ2	74.0	0.0	-18.8	54.0	440.3	769.4	0.0	-18.8	54.0	440.3

L/C	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201 1.0DL+1.0LL	1205.4	0.1	-228.1	-30.0	304.5	10.6	11.9	4821.6	16.1
202 1.0DL+0.75LL	1112.9	0.1	-184.5	-25.0	247.6	12.1	13.5	4451.6	17.8
203 1.0DL+1.0WLX1	845.4	-100.0	776.5	-10.0	101.5	2.2	25.0	3.4	33.8
204 1.0DL+1.0WLX2	825.4	100.0	-883.9	-10.0	52.5	1.9	47.2	3.3	33.0
205 1.0DL+1.0WLZ1	845.4	0.0	-67.1	-70.0	563.5	25.2	4.5	NA	4.8
206 1.0DL+1.0WLZ2	825.4	0.0	-40.3	50.0	409.5	41.0	6.0	NA	6.6
207 1.0DL+0.75LL+0.75WLX1	1120.4	-75.0	438.1	-25.0	266.0	5.1	12.6	6.0	17.9
208 1.0DL+0.75LL+0.75WLX2	1105.4	75.0	-807.2	-25.0	229.3	2.7	14.5	5.9	17.7
209 1.0DL+0.75LL+0.75WLZ1	1120.4	0.0	-194.6	-70.0	612.5	11.5	5.5	NA	6.4
210 1.0DL+0.75LL+0.75WLZ2	1105.4	0.0	-174.5	20.0	117.3	12.7	28.3	NA	22.1
211 0.6DL+1.0WLX1	789.4	-100.0	798.0	-6.0	70.7	2.0	33.5	3.2	52.6
212 0.6DL+1.0WLX2	769.4	100.0	-862.4	-6.0	21.7	1.8	106.4	3.1	51.3
213 0.6DL+1.0WLZ1	789.4	0.0	-45.6	-66.0	532.7	34.6	4.4	NA	4.8
214 0.6DL+1.0WLZ2	769.4	0.0	-18.8	54.0	440.3	81.9	5.2	NA	5.7
					Min.=	**1.8**	**4.4**	**3.1**	**4.8**

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Base pressure check of footing F2 :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	84.139	0
Min Px	1	201	0	84.139	0
Max Py	16	208	0	194.32	0
Min Py	7	203	0	0	0
Max Pz	1	201	0	84.139	0
Min Pz	1	201	0	84.139	0

Gross Maximum Base pressure, p_{max} = 194.3 KN/m^2

Net Allowable Safe Bearing Capacity = 500 KN/m^2 (Bore hole 26)
Depth of soil above footing = 1.1 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 519.8 KN/m^2
> 194.3 KN/m^2
Hence OK

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Design of Foundation Raft F2 :

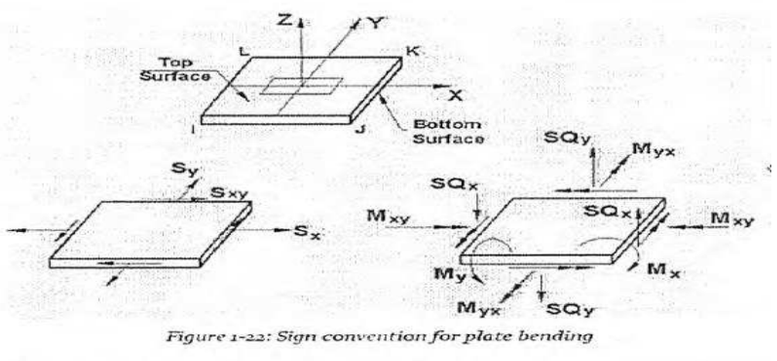
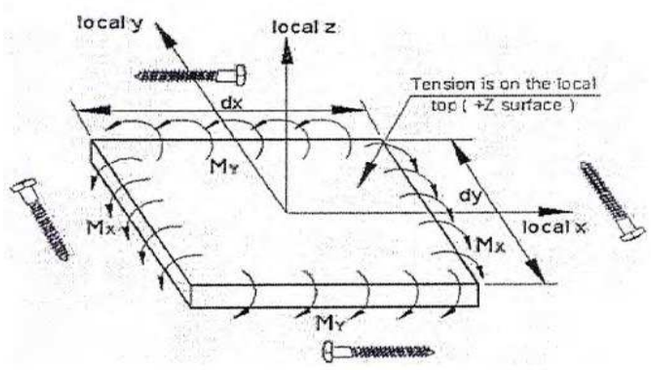


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; : FDN_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	271	0.876	-0.113	0	0	0	-24.03	-7.903	30.707	-54.733	-38.61	102 1.4DL+1.7LL
Min Qx	541	-0.702	0.083	0	0	0	-50.38	-27.593	0	-50.378	-27.593	104 1.2DL+1.0LL+1.6WLX2
Max Qy	548	-0.053	0.773	0	0	0	-3.784	30.799	21.364	-25.148	52.163	104 1.2DL+1.0LL+1.6WLX2
Min Qy	582	-0.059	-0.518	0	0	0	-2.919	3.735	-13.125	-16.044	16.86	104 1.2DL+1.0LL+1.6WLX2
Max Sx	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Min Sx	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Max Sy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Min Sy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Max Sxy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Min Sxy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Max Mx	308	-0.028	0.11	0	0	0	59.502	-3.441	-14.787	74.289	-18.228	103 1.2DL+1.0LL+1.6WLX1
Min Mx	58	0.024	-0.389	0	0	0	-62.3	-20.863	14.667	-76.964	-35.53	102 1.4DL+1.7LL
Max My	119	0.112	-0.036	0	0	0	-7.421	101.906	16.892	-24.313	118.798	102 1.4DL+1.7LL
Min My	187	-0.091	-0.284	0	0	0	-12.23	-74.312	2.945	-15.178	-77.257	109 0.9DL+1.6HL+1.6WLZ1
Max Mxy	322	-0.125	-0.077	0	0	0	40.569	53.987	69.621	110.19	123.608	104 1.2DL+1.0LL+1.6WLX2
Min Mxy	509	0.034	-0.02	0	0	0	22.954	19.383	-44.184	67.138	63.567	104 1.2DL+1.0LL+1.6WLX2

PLATE CENTER STRESS ; FDN_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	273	0.671	0.073	0	0	0	15.226	1.125	30.223	45.449	31.348	102 1.4DL+1.7LL
Min Qx	96	-0.248	0.068	0	0	0	-4.596	-3.898	0	-4.596	-3.898	109 0.9DL+1.6HL+1.6WLZ1
Max Qy	548	-0.053	0.773	0	0	0	-3.784	30.799	21.364	-25.148	52.163	104 1.2DL+1.0LL+1.6WLX2
Min Qy	582	-0.059	-0.518	0	0	0	-2.919	3.735	-13.125	-16.044	16.86	104 1.2DL+1.0LL+1.6WLX2
Max Sx	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Min Sx	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Max Sy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Min Sy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Max Sxy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Min Sxy	625	0.01	-0.023	0	0	0	-0.159	-0.97	0.086	-0.245	-1.056	101 1.4DL
Max Mx	306	-0.009	0.029	0	0	0	53.874	6.184	-17.681	71.555	23.865	103 1.2DL+1.0LL+1.6WLX1
Min Mx	294	0.151	0.009	0	0	0	-29.95	-1.195	-18.419	-48.366	-19.614	107 0.9DL+1.6HL+1.6WLX1
Max My	123	0.018	0.007	0	0	0	4.446	90.718	21.545	25.991	112.263	102 1.4DL+1.7LL
Min My	582	-0.045	0.087	0	0	0	-0.4	-53.001	2.297	-2.697	-55.298	109 0.9DL+1.6HL+1.6WLZ1
Max Mxy	312	-0.034	0.1	0	0	0	43.355	27.6	57.344	100.699	84.944	104 1.2DL+1.0LL+1.6WLX2
Min Mxy	342	-0.087	0.088	0	0	0	14.576	13.691	-38.051	52.627	51.742	107 0.9DL+1.6HL+1.6WLX1

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

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Reinforcement Along X:

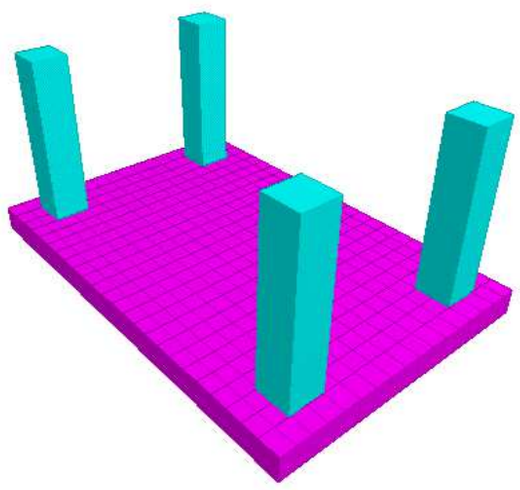
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	74.29	top face	50	342	74.289	0.85	0.0020	699.7	720	16	279.37	150	1341	18.91	23.64	0.04	tension control
Min Mx	-76.96	bottom face	50	342	76.964	0.88	0.0021	725.3		16	277.33	150	1341	18.91	23.64	0.04	tension control
Max Mxy	110.19	top face	50	342	110.19	1.26	0.0031	1045.4		16	192.41	150	1341	18.91	23.64	0.04	tension control
Min Mxy	67.14	top face	50	342	67.138	0.77	0.0018	631.4		16	279.37	150	1341	18.91	23.64	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.671	1	268.4	45.449	0.004	0.342	369.771	346.525	586.757	346.52	1261	> 268.4	ok	259.9	φVc<Vu	required
	Min Qx	0.248	1	99.2	4.596			492.009			346.52	1261	> 99.2	ok	259.9	φVc>Vu	not required

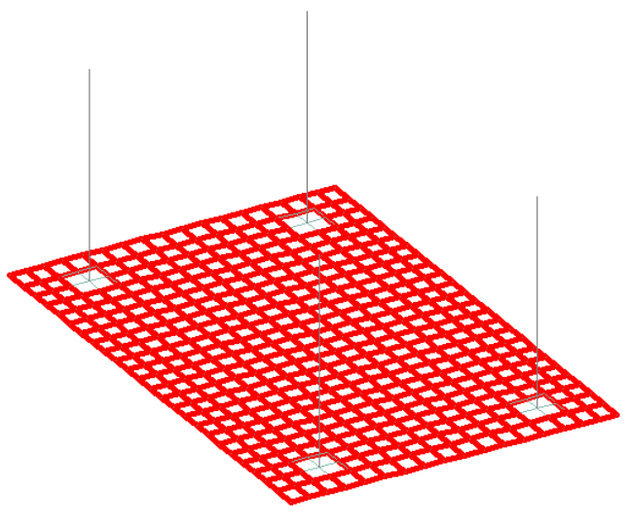
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	118.80	top face	50	342	118.8	1.35	0.0033	1129.0	720	16	178.15	150	1341	18.91	23.64	0.04	tension control
Min My	-77.26	bottom face	50	342	77.257	0.88	0.0021	728.1		16	276.27	150	1341	18.91	23.64	0.04	tension control
Max Mxy	123.61	top face	50	342	123.61	1.41	0.0034	1175.9		16	171.05	150	1341	18.91	23.64	0.04	tension control
Min Mxy	63.57	top face	50	342	63.567	0.72	0.0017	597.4		16	279.37	150	1341	18.91	23.64	0.04	tension control

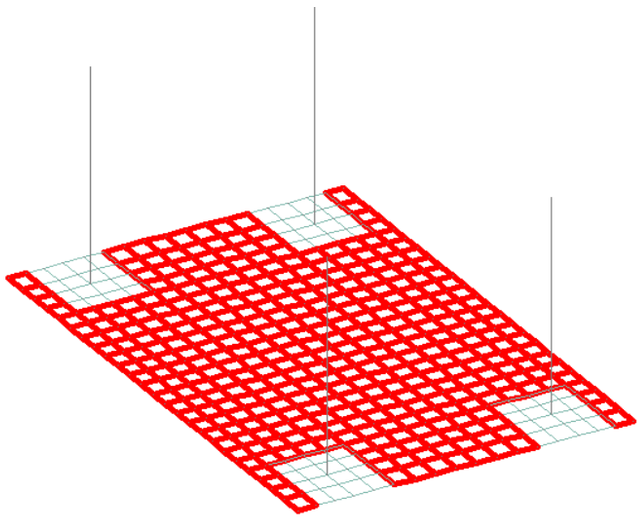
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.773	1	309.2	52.163	0.004	0.342	fc'369.9	fc'346.5	fc'586.8	346.52	360	> 309.2	ok	259.9	φVc<Vu	required
	Min Qy	0.518	1	207.2	16.86			419.543			346.52	360	> 207.2	ok	259.9	φVc>Vu	not required



3D Rendered View of foundation



Highlighted plate for moment (at column face)



Highlighted plate for shear (at d distance from column face)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	**150**	c/c
At Top	16ϕT	**150**	c/c
Along Z Direction :			
At Bottom	16ϕT	**150**	c/c
At Top	16ϕT	**150**	c/c

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5.3 DESIGN OF COLUMN C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 2.4 m kl_{ux}/r_x = 19.2 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.4 m kl_{uz}/r_z = 19.2

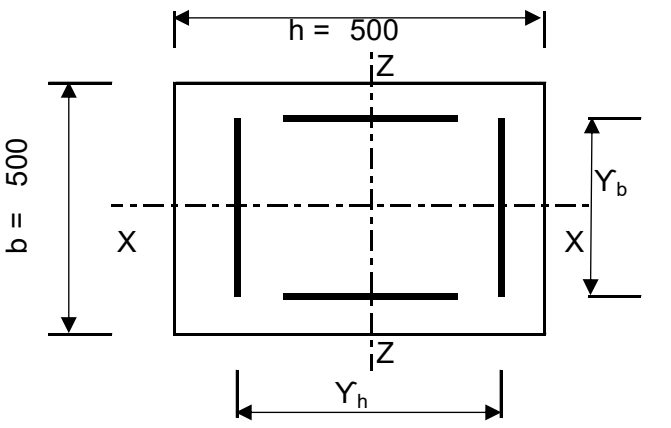
Radius of gyration r_x= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845.08 KN safe

Radius of gyration r_z= 150 mm

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx}=M_{ux}/φ$	$M_{nz}=M_{uz}/φ$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	200.8	44.2	0.0	309	68.0	0.0	0.0353	0.050	218.75	0.31	0.050	218.8	0.00	1.609	0.153	safe
109	30.0	0.0	0.0	46	0.0	0.0	0.0053	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
105	35.5	57.2	0.0	55	88.0	0.0	0.0062	0.050	218.75	0.40	0.050	218.8	0.00		0.232	safe
109	89.6	83.2	0.0	138	128.0	0.0	0.0158	0.050	218.75	0.59	0.050	218.8	0.00		0.423	safe
106	131.5	109.2	0.0	202	168.0	0.0	0.0231	0.050	218.75	0.77	0.050	218.8	0.00		0.654	safe
105	183.5	109.2	0.0	282	168.0	0.0	0.0323	0.050	218.75	0.77	0.050	218.8	0.00		0.654	safe
101	50.8	0.0	0.0	78	0.0	0.0	0.0089	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
101	50.8	0.0	0.0	78	0.0	0.0	0.0089	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
105	183.5	109.2	0.0	282	168.0	0.0	0.0323	0.050	218.75	0.77	0.050	218.8	0.00		0.654	safe
106	131.5	109.2	0.0	202	168.0	0.0	0.0231	0.050	218.75	0.77	0.050	218.8	0.00		0.654	safe
105	35.5	57.2	0.0	55	88.0	0.0	0.0062	0.050	218.75	0.40	0.050	218.8	0.00		0.232	safe
109	89.6	83.2	0.0	138	128.0	0.0	0.0158	0.050	218.75	0.59	0.050	218.8	0.00		0.423	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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5.4 DESIGN OF COLUMN C2

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.5 % i.e. ρ_g = 0.015

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 2.4 m kl_{ux}/r_x = 19.2 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.4 m kl_{uz}/r_z = 19.2

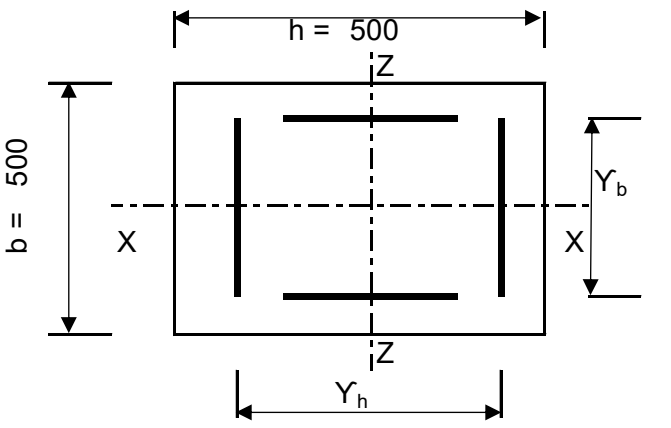
Radius of gyration r_x= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 7127.62 KN safe

Radius of gyration r_z= 150 mm

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx}=M_{ux}/\varphi$	$M_{nz}=M_{uz}/\varphi$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha=\log0.5/\log\beta$	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
104	395.5	109.2	124.8	608	168.0	192.0	0.0695	0.080	350	0.48	0.080	350	0.55	1.609	0.688	safe
108	165.0	0.0	0.0	254	0.0	0.0	0.0290	0.070	306.25	0.00	0.070	306.3	0.00		0	safe
103	43.5	109.2	124.8	67	168.0	192.0	0.0076	0.070	306.25	0.55	0.070	306.3	0.63		0.853	safe
104	395.5	109.2	124.8	608	168.0	192.0	0.0695	0.080	350	0.48	0.080	350	0.55		0.688	safe
106	225.5	150.8	0.0	347	232.0	0.0	0.0396	0.080	350	0.66	0.080	350	0.00		0.517	safe
102	330.8	169.0	0.0	509	260.0	0.0	0.0582	0.080	350	0.74	0.080	350	0.00		0.62	safe
101	78.8	0.0	0.0	121	0.0	0.0	0.0139	0.070	306.25	0.00	0.070	306.3	0.00		0	safe
101	78.8	0.0	0.0	121	0.0	0.0	0.0139	0.070	306.25	0.00	0.070	306.3	0.00		0	safe
102	330.8	169.0	0.0	509	260.0	0.0	0.0582	0.080	350	0.74	0.080	350	0.00		0.62	safe
106	225.5	150.8	0.0	347	232.0	0.0	0.0396	0.080	350	0.66	0.080	350	0.00		0.517	safe
103	43.5	109.2	124.8	67	168.0	192.0	0.0076	0.070	306.25	0.55	0.070	306.3	0.63		0.853	safe
104	395.5	109.2	124.8	608	168.0	192.0	0.0695	0.080	350	0.48	0.080	350	0.55		0.688	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3750	4	20	+	8	20	3772	1.51	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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ATTACHMENT 1 STAAD INPUT & OUTPUT FILE FOR F1 DESIGN

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR  8, 2020                   *
*           Time=    11:33:49                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F1 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 -0.55; 2 0 0 5.45; 3 0 0 0; 4 0 0 4.9; 5 0.75 0 -0.55; 6 0.75 0 5.45
     9. 7 0.75 0 0; 8 0.75 0 4.9; 9 -0.75 0 -0.55; 10 -0.75 0 5.45; 11 -0.75 0 0
    10. 12 -0.75 0 4.9; 13 0 2.8 0; 14 0 2.8 4.9; 15 -0.5 0 0; 16 -0.5 0 -0.25
    11. 17 -0.75 0 -0.25; 18 -0.25 0 0; 19 -0.25 0 -0.25; 20 0 0 -0.25
    12. 21 -0.5 0 -0.55; 22 -0.25 0 -0.55; 23 0.25 0 0; 24 0.25 0 -0.25; 25 0.5 0 0
    13. 26 0.5 0 -0.25; 27 0.75 0 -0.25; 28 0.25 0 -0.55; 29 0.5 0 -0.55
    14. 30 -0.5 0 5.45; 31 -0.5 0 5.15; 32 -0.75 0 5.15; 33 -0.25 0 5.45
    15. 34 -0.25 0 5.15; 35 0 0 5.15; 36 -0.5 0 4.9; 37 -0.25 0 4.9; 38 0.25 0 5.45
    16. 39 0.25 0 5.15; 40 0.5 0 5.45; 41 0.5 0 5.15; 42 0.75 0 5.15; 43 0.25 0 4.9
    17. 44 0.5 0 4.9; 45 -0.5 0 4.655; 46 -0.75 0 4.655; 47 -0.25 0 4.655
    18. 48 0 0 4.655; 49 -0.5 0 4.41; 50 -0.75 0 4.41; 51 -0.25 0 4.41; 52 0 0 4.41
    19. 53 -0.5 0 4.165; 54 -0.75 0 4.165; 55 -0.25 0 4.165; 56 0 0 4.165
    20. 57 -0.5 0 3.92; 58 -0.75 0 3.92; 59 -0.25 0 3.92; 60 0 0 3.92; 61 -0.5 0 3.675
    21. 62 -0.75 0 3.675; 63 -0.25 0 3.675; 64 0 0 3.675; 65 -0.5 0 3.43
    22. 66 -0.75 0 3.43; 67 -0.25 0 3.43; 68 0 0 3.43; 69 -0.5 0 3.185
    23. 70 -0.75 0 3.185; 71 -0.25 0 3.185; 72 0 0 3.185; 73 -0.5 0 2.94
    24. 74 -0.75 0 2.94; 75 -0.25 0 2.94; 76 0 0 2.94; 77 -0.5 0 2.695
    25. 78 -0.75 0 2.695; 79 -0.25 0 2.695; 80 0 0 2.695; 81 -0.5 0 2.45
    26. 82 -0.75 0 2.45; 83 -0.25 0 2.45; 84 0 0 2.45; 85 -0.5 0 2.205
    27. 86 -0.75 0 2.205; 87 -0.25 0 2.205; 88 0 0 2.205; 89 -0.5 0 1.96
    28. 90 -0.75 0 1.96; 91 -0.25 0 1.96; 92 0 0 1.96; 93 -0.5 0 1.715
    29. 94 -0.75 0 1.715; 95 -0.25 0 1.715; 96 0 0 1.715; 97 -0.5 0 1.47
    30. 98 -0.75 0 1.47; 99 -0.25 0 1.47; 100 0 0 1.47; 101 -0.5 0 1.225
    31. 102 -0.75 0 1.225; 103 -0.25 0 1.225; 104 0 0 1.225; 105 -0.5 0 0.979999
    32. 106 -0.75 0 0.979999; 107 -0.25 0 0.979999; 108 0 0 0.979999
    33. 109 -0.5 0 0.734999; 110 -0.75 0 0.734999; 111 -0.25 0 0.734999
    34. 112 0 0 0.734999; 113 -0.5 0 0.489999; 114 -0.75 0 0.489999
    35. 115 -0.25 0 0.489999; 116 0 0 0.489999; 117 -0.5 0 0.244999
    36. 118 -0.75 0 0.244999; 119 -0.25 0 0.244999; 120 0 0 0.244999; 121 0.25 0 4.655
    37. 122 0.5 0 4.655; 123 0.75 0 4.655; 124 0.25 0 4.41; 125 0.5 0 4.41
    38. 126 0.75 0 4.41; 127 0.25 0 4.165; 128 0.5 0 4.165; 129 0.75 0 4.165
```

```
39. 130 0.25 0 3.92; 131 0.5 0 3.92; 132 0.75 0 3.92; 133 0.25 0 3.675
40. 134 0.5 0 3.675; 135 0.75 0 3.675; 136 0.25 0 3.43; 137 0.5 0 3.43
41. 138 0.75 0 3.43; 139 0.25 0 3.185; 140 0.5 0 3.185; 141 0.75 0 3.185
42. 142 0.25 0 2.94; 143 0.5 0 2.94; 144 0.75 0 2.94; 145 0.25 0 2.695
43. 146 0.5 0 2.695; 147 0.75 0 2.695; 148 0.25 0 2.45; 149 0.5 0 2.45
44. 150 0.75 0 2.45; 151 0.25 0 2.205; 152 0.5 0 2.205; 153 0.75 0 2.205
45. 154 0.25 0 1.96; 155 0.5 0 1.96; 156 0.75 0 1.96; 157 0.25 0 1.715
46. 158 0.5 0 1.715; 159 0.75 0 1.715; 160 0.25 0 1.47; 161 0.5 0 1.47
47. 162 0.75 0 1.47; 163 0.25 0 1.225; 164 0.5 0 1.225; 165 0.75 0 1.225
48. 166 0.25 0 0.979999; 167 0.5 0 0.979999; 168 0.75 0 0.979999
49. 169 0.25 0 0.734999; 170 0.5 0 0.734999; 171 0.75 0 0.734999
50. 172 0.25 0 0.489999; 173 0.5 0 0.489999; 174 0.75 0 0.489999
51. 175 0.25 0 0.244999; 176 0.5 0 0.244999; 177 0.75 0 0.244999; 178 0 0.2 0
52. 179 0 0.2 4.9
53. MEMBER INCIDENCES
54. 18 3 178; 19 4 179; 249 178 13; 250 179 14
55. ELEMENT INCIDENCES SHELL
56. 28 11 15 16 17; 30 15 18 19 16; 32 18 3 20 19; 34 17 16 21 9; 36 16 19 22 21
57. 37 19 20 1 22; 39 3 23 24 20; 41 23 25 26 24; 43 25 7 27 26; 45 20 24 28 1
58. 47 24 26 29 28; 48 26 27 5 29; 51 10 30 31 32; 53 30 33 34 31; 55 33 2 35 34
59. 57 32 31 36 12; 59 31 34 37 36; 60 34 35 4 37; 62 2 38 39 35; 64 38 40 41 39
60. 66 40 6 42 41; 68 35 39 43 4; 70 39 41 44 43; 71 41 42 8 44; 73 12 36 45 46
61. 74 36 37 47 45; 76 37 4 48 47; 78 46 45 49 50; 79 45 47 51 49; 81 47 48 52 51
62. 83 50 49 53 54; 84 49 51 55 53; 86 51 52 56 55; 88 54 53 57 58; 89 53 55 59 57
63. 91 55 56 60 59; 93 58 57 61 62; 94 57 59 63 61; 96 59 60 64 63; 98 62 61 65 66
64. 99 61 63 67 65; 101 63 64 68 67; 103 66 65 69 70; 104 65 67 71 69
65. 106 67 68 72 71; 108 70 69 73 74; 109 69 71 75 73; 111 71 72 76 75
66. 113 74 73 77 78; 114 73 75 79 77; 116 75 76 80 79; 118 78 77 81 82
67. 119 77 79 83 81; 121 79 80 84 83; 123 82 81 85 86; 124 81 83 87 85
68. 126 83 84 88 87; 128 86 85 89 90; 129 85 87 91 89; 131 87 88 92 91
69. 133 90 89 93 94; 134 89 91 95 93; 136 91 92 96 95; 138 94 93 97 98
70. 139 93 95 99 97; 141 95 96 100 99; 143 98 97 101 102; 144 97 99 103 101
71. 146 99 100 104 103; 148 102 101 105 106; 149 101 103 107 105
72. 151 103 104 108 107; 153 106 105 109 110; 154 105 107 111 109
73. 156 107 108 112 111; 158 110 109 113 114; 159 109 111 115 113
74. 161 111 112 116 115; 163 114 113 117 118; 164 113 115 119 117
75. 166 115 116 120 119; 167 118 117 15 11; 168 117 119 18 15; 169 119 120 3 18
76. 170 4 43 121 48; 171 43 44 122 121; 173 44 8 123 122; 174 48 121 124 52
77. 175 121 122 125 124; 177 122 123 126 125; 178 52 124 127 56
78. 179 124 125 128 127; 181 125 126 129 128; 182 56 127 130 60
79. 183 127 128 131 130; 185 128 129 132 131; 186 60 130 133 64
80. 187 130 131 134 133; 189 131 132 135 134; 190 64 133 136 68
81. 191 133 134 137 136; 193 134 135 138 137; 194 68 136 139 72
82. 195 136 137 140 139; 197 137 138 141 140; 198 72 139 142 76
83. 199 139 140 143 142; 201 140 141 144 143; 202 76 142 145 80
84. 203 142 143 146 145; 205 143 144 147 146; 206 80 145 148 84
85. 207 145 146 149 148; 209 146 147 150 149; 210 84 148 151 88
86. 211 148 149 152 151; 213 149 150 153 152; 214 88 151 154 92
87. 215 151 152 155 154; 217 152 153 156 155; 218 92 154 157 96
88. 219 154 155 158 157; 221 155 156 159 158; 222 96 157 160 100
89. 223 157 158 161 160; 225 158 159 162 161; 226 100 160 163 104
90. 227 160 161 164 163; 229 161 162 165 164; 230 104 163 166 108
91. 231 163 164 167 166; 233 164 165 168 167; 234 108 166 169 112
92. 235 166 167 170 169; 237 167 168 171 170; 238 112 169 172 116
93. 239 169 170 173 172; 241 170 171 174 173; 242 116 172 175 120
94. 243 172 173 176 175; 245 173 174 177 176; 246 120 175 23 3; 247 175 176 25 23
```

```
 95. 248 176 177 7 25
 96. START GROUP DEFINITION
 97. ELEMENT
 98. _RAFT 28 30 32 34 36 37 39 41 43 45 47 48 51 53 55 57 59 60 62 64 66 68 70 -
 99. 71 73 74 76 78 79 81 83 84 86 88 89 91 93 94 96 98 99 101 103 104 106 108 -
100. 109 111 113 114 116 118 119 121 123 124 126 128 129 131 133 134 136 138 139 -
101. 141 143 144 146 148 149 151 153 154 156 158 159 161 163 164 166 TO 171 173 -
102. 174 TO 175 177 TO 179 181 TO 183 185 TO 187 189 TO 191 193 TO 195 197 TO 199 -
103. 201 TO 203 205 TO 207 209 TO 211 213 TO 215 217 TO 219 221 TO 223 -
104. 225 TO 227 229 TO 231 233 TO 235 237 TO 239 241 TO 243 245 TO 248
105. _RAFT_MOM 28 30 34 36 37 41 43 45 47 48 51 53 55 57 59 62 64 66 70 71 73 74 -
106. 78 79 81 83 84 86 88 89 91 93 94 96 98 99 101 103 104 106 108 109 111 113 -
107. 114 116 118 119 121 123 124 126 128 129 131 133 134 136 138 139 141 143 144 -
108. 146 148 149 151 153 154 156 158 159 161 163 164 166 TO 168 171 173 TO 175 -
109. 177 TO 179 181 TO 183 185 TO 187 189 TO 191 193 TO 195 197 TO 199 -
110. 201 TO 203 205 TO 207 209 TO 211 213 TO 215 217 TO 219 221 TO 223 -
111. 225 TO 227 229 TO 231 233 TO 235 237 TO 239 241 TO 243 245 247 248
112. _RAFT_SHEAR 28 34 43 48 51 57 66 71 73 78 83 84 86 88 89 91 93 94 96 98 99 -
113. 101 103 104 106 108 109 111 113 114 116 118 119 121 123 124 126 128 129 131 -
114. 133 134 136 138 139 141 143 144 146 148 149 151 153 154 156 158 159 161 163 -
115. 167 173 177 TO 179 181 TO 183 185 TO 187 189 TO 191 193 TO 195 197 TO 199 -
116. 201 TO 203 205 TO 207 209 TO 211 213 TO 215 217 TO 219 221 TO 223 -
117. 225 TO 227 229 TO 231 233 TO 235 237 TO 239 241 245 248
118. MEMBER
119. _C1 249 250
120. END GROUP DEFINITION
121. ELEMENT PROPERTY
122. 28 30 32 34 36 37 39 41 43 45 47 48 51 53 55 57 59 60 62 64 66 68 70 71 73 -
123. 74 76 78 79 81 83 84 86 88 89 91 93 94 96 98 99 101 103 104 106 108 109 111 -
124. 113 114 116 118 119 121 123 124 126 128 129 131 133 134 136 138 139 141 143 -
125. 144 146 148 149 151 153 154 156 158 159 161 163 164 166 TO 171 173 TO 175 -
126. 177 TO 179 181 TO 183 185 TO 187 189 TO 191 193 TO 195 197 TO 199 -
127. 201 TO 203 205 TO 207 209 TO 211 213 TO 215 217 TO 219 221 TO 223 -
128. 225 TO 227 229 TO 231 233 TO 235 237 TO 239 241 TO 243 245 TO 247 -
129. 248 THICKNESS 0.4
130. DEFINE MATERIAL START
131. ISOTROPIC CONCRETE
132. E 2.17185E+007
133. POISSON 0.17
134. DENSITY 25
135. ALPHA 1E-005
136. DAMP 0.05
137. TYPE CONCRETE
138. STRENGTH FCU 27579
139. END DEFINE MATERIAL
140. MEMBER PROPERTY AMERICAN
141. 18 19 249 250 PRIS YD 0.5 ZD 0.5
142. CONSTANTS
143. MATERIAL CONCRETE ALL
144. **********************************************************
145. SUPPORTS
146. 28 30 32 34 36 37 39 41 43 45 47 48 51 53 55 57 59 60 62 64 66 68 70 71 73 -
147. 74 76 78 79 81 83 84 86 88 89 91 93 94 96 98 99 101 103 104 106 108 109 111 -
148. 113 114 116 118 119 121 123 124 126 128 129 131 133 134 136 138 139 141 143 -
149. 144 146 148 149 151 153 154 156 158 159 161 163 164 166 TO 171 173 TO 175 -
150. 177 TO 179 181 TO 183 185 TO 187 189 TO 191 193 TO 195 197 TO 199 -
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151. 201 TO 203 205 TO 207 209 TO 211 213 TO 215 217 TO 219 221 TO 223 -
152. 225 TO 227 229 TO 231 233 TO 235 237 TO 239 241 TO 243 245 TO 247 -
153. 248 PLATE MAT DIRECT Y SUBGRADE 60000 PRINT
```

JOINT	INFLUENCE AREA
1	0.037500
2	0.037500
3	0.061875
4	0.061875
5	0.018750
6	0.018750
7	0.030937
8	0.030938
9	0.018750
10	0.018750
11	0.030937
12	0.030938
15	0.061875
16	0.068750
17	0.034375
18	0.061875
19	0.068750
20	0.068750
21	0.037500
22	0.037500
23	0.061875
24	0.068750
25	0.061875
26	0.068750
27	0.034375
28	0.037500
29	0.037500
30	0.037500
31	0.068750
32	0.034375
33	0.037500
34	0.068750
35	0.068750
36	0.061875
37	0.061875
38	0.037500
39	0.068750
40	0.037500
41	0.068750
42	0.034375
43	0.061875
44	0.061875
45	0.061250
46	0.030625
47	0.061250
48	0.061250
49	0.061250
50	0.030625
51	0.061250

52	0.061250
53	0.061250
54	0.030625
55	0.061250
56	0.061250
57	0.061250
58	0.030625
59	0.061250
60	0.061250
61	0.061250
62	0.030625
63	0.061250
64	0.061250
65	0.061250
66	0.030625
67	0.061250
68	0.061250
69	0.061250
70	0.030625
71	0.061250
72	0.061250
73	0.061250
74	0.030625
75	0.061250
76	0.061250
77	0.061250
78	0.030625
79	0.061250
80	0.061250
81	0.061250
82	0.030625
83	0.061250
84	0.061250
85	0.061250
86	0.030625
87	0.061250
88	0.061250
89	0.061250
90	0.030625
91	0.061250
92	0.061250
93	0.061250
94	0.030625
95	0.061250
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97	0.061250
98	0.030625
99	0.061250
100	0.061250
101	0.061250
102	0.030625
103	0.061250
104	0.061250
105	0.061250
106	0.030625
107	0.061250

108	0.061250
109	0.061250
110	0.030625
111	0.061250
112	0.061250
113	0.061250
114	0.030625
115	0.061250
116	0.061250
117	0.061250
118	0.030625
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126	0.030625
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130	0.061250
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132	0.030625
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136	0.061250
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139	0.061250
140	0.061250
141	0.030625
142	0.061250
143	0.061250
144	0.030625
145	0.061250
146	0.061250
147	0.030625
148	0.061250
149	0.061250
150	0.030625
151	0.061250
152	0.061250
153	0.030625
154	0.061250
155	0.061250
156	0.030625
157	0.061250
158	0.061250
159	0.030625
160	0.061250
161	0.061250
162	0.030625
163	0.061250

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       164          0.061250
       165          0.030625
       166          0.061250
       167          0.061250
       168          0.030625
       169          0.061250
       170          0.061250
       171          0.030625
       172          0.061250
       173          0.061250
       174          0.030625
       175          0.061250
       176          0.061250
       177          0.030625
    TOTAL           9.000006
154. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
155. JOINT LOAD
156. 13 FY -20
157. 14 FY -30
158. SELFWEIGHT Y -1
159. ***
160. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
161. JOINT LOAD
162. 13 FY -40 FZ -10
163. 14 FY -80 FZ 10
164. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
165. JOINT LOAD
166. 13 FX 0 FY 0 FZ 0
167. 14 FX 0 FY 0 FZ 0
168. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
169. JOINT LOAD
170. 13 FX 0 FY 0 FZ 0
171. 14 FX 0 FY 0 FZ 0
172. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
173. JOINT LOAD
174. 13 FY 30 FZ 20
175. 14 FY -30 FZ 20
176. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
177. JOINT LOAD
178. 13 FY -30 FZ -20
179. 14 FY 30 FZ -20
180. LOAD 7 LOADTYPE SOIL  TITLE SOIL LOAD (HL)
181. ELEMENT LOAD
182. 28 30 32 34 36 37 39 41 43 45 47 48 51 53 55 57 59 60 62 64 66 68 70 71 73 -
183. 74 76 78 79 81 83 84 86 88 89 91 93 94 96 98 99 101 103 104 106 108 109 111 -
184. 113 114 116 118 119 121 123 124 126 128 129 131 133 134 136 138 139 141 143 -
185. 144 146 148 149 151 153 154 156 158 159 161 163 164 166 TO 171 173 TO 175 -
186. 177 TO 179 181 TO 183 185 TO 187 189 TO 191 193 TO 195 197 TO 199 -
187. 201 TO 203 205 TO 207 209 TO 211 213 TO 215 217 TO 219 221 TO 223 -
188. 225 TO 227 229 TO 231 233 TO 235 237 TO 239 241 TO 243 245 TO 248 PR -19.8
189. ************************************************
190. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
191. LOAD COMB 101 1.4DL
192. 1 1.4
193. LOAD COMB 102 1.4DL+1.7LL
194. 1 1.4 2 1.7
```

```
195. LOAD COMB 103 1.2DL+1.0LL+1.6WLX1
196. 1 1.2 2 1.0 3 1.6
197. LOAD COMB 104 1.2DL+1.0LL+1.6WLX2
198. 1 1.2 2 1.0 4 1.6
199. LOAD COMB 105 1.2DL+1.0LL+1.6WLZ1
200. 1 1.2 2 1.0 5 1.6
201. LOAD COMB 106 1.2DL+1.0LL+1.6WLZ2
202. 1 1.2 2 1.0 6 1.6
203. LOAD COMB 107 0.9DL+1.6HL+1.6WLX1
204. 1 0.9 7 1.6 3 1.6
205. LOAD COMB 108 0.9DL+1.6HL+1.6WLX2
206. 1 0.9 7 1.6 4 1.6
207. LOAD COMB 109 0.9DL+1.6HL+1.6WLZ1
208. 1 0.9 7 1.6 5 1.6
209. LOAD COMB 110 0.9DL+1.6HL+1.6WLZ2
210. 1 0.9 7 1.6 6 1.6
211. ***************
212. ********************************************************************************
213. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
214. ************************************************************
215. LOAD COMB 201 1.0DL+1.0HL+1.0LL
216. 1 1.0 7 1.0 2 1.0
217. LOAD COMB 202 1.0DL+1.0HL+0.75LL
218. 1 1.0 7 1.0 2 0.75
219. LOAD COMB 203 1.0DL+1.0HL+1.0WLX1
220. 1 1.0 7 1.0 3 1.0
221. LOAD COMB 204 1.0DL+1.0HL+1.0WLX2
222. 1 1.0 7 1.0 4 1.0
223. LOAD COMB 205 1.0DL+1.0HL+1.0WLZ1
224. 1 1.0 7 1.0 5 1.0
225. LOAD COMB 206 1.0DL+1.0HL+1.0WLZ2
226. 1 1.0 7 1.0 6 1.0
227. LOAD COMB 207 1.0DL+1.0HL+0.75LL+0.75WLX1
228. 1 1.0 7 1.0 2 0.75 3 0.75
229. LOAD COMB 208 1.0DL+1.0HL+0.75LL+0.75WLX2
230. 1 1.0 7 1.0 2 0.75 4 0.75
231. LOAD COMB 209 1.0DL+1.0HL+0.75LL+0.75WLZ1
232. 1 1.0 7 1.0 2 0.75 5 0.75
233. LOAD COMB 210 1.0DL+1.0HL+0.75LL+0.75WLZ2
234. 1 1.0 7 1.0 2 0.75 6 0.75
235. LOAD COMB 211 0.6DL+1.0HL+1.0WLX1
236. 1 0.6 7 1.0 3 1.0
237. LOAD COMB 212 0.6DL+1.0HL+1.0WLX2
238. 1 0.6 7 1.0 4 1.0
239. LOAD COMB 213 0.6DL+1.0HL+1.0WLZ1
240. 1 0.6 7 1.0 5 1.0
241. LOAD COMB 214 0.6DL+1.0HL+1.0WLX1
242. 1 0.6 7 1.0 6 1.0
243. ********
244. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 179 NUMBER OF MEMBERS 4
NUMBER OF PLATES 144 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 175

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 175/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 549
TOTAL LOAD COMBINATION CASES = 24 SO FAR.
SIZE OF STIFFNESS MATRIX = 25 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.1/ 112067.5 MB

245. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 8,2020 TIME= 11:33:56 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 2 STAAD INPUT & OUTPUT FILE FOR F1 STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR  8, 2020                   *
*           Time=    11:36:41                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F1 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 3 0 0 0; 4 0 0 4.9; 13 0 2.8 0; 14 0 2.8 4.9
     9. MEMBER INCIDENCES
    10. 18 3 13; 19 4 14
    11. DEFINE MATERIAL START
    12. ISOTROPIC CONCRETE
    13. E 2.17185E+007
    14. POISSON 0.17
    15. DENSITY 25
    16. ALPHA 1E-005
    17. DAMP 0.05
    18. TYPE CONCRETE
    19. STRENGTH FCU 27579
    20. END DEFINE MATERIAL
    21. MEMBER PROPERTY AMERICAN
    22. 18 19 PRIS YD 0.5 ZD 0.5
    23. CONSTANTS
    24. MATERIAL CONCRETE ALL
    25. ***********************************************************
    26. SUPPORTS
    27. 3 4 FIXED
    28. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    29. JOINT LOAD
    30. 13 FY -20
    31. 14 FY -30
    32. ***
    33. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
    34. JOINT LOAD
    35. 13 FY -40 FZ -10
    36. 14 FY -80 FZ 10
```

```
37. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
38. JOINT LOAD
39. 13 FX 0 FY 0 FZ 0
40. 14 FX 0 FY 0 FZ 0
41. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
42. JOINT LOAD
43. 13 FX 0 FY 0 FZ 0
44. 14 FX 0 FY 0 FZ 0
45. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
46. JOINT LOAD
47. 13 FY 30 FZ 20
48. 14 FY -30 FZ 20
49. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
50. JOINT LOAD
51. 13 FY -30 FZ -20
52. 14 FY 30 FZ -20
53. ****************************************************
54. ********************************************************************************
55. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
56. ************************************************************
57. LOAD COMB 201 1.0DL+1.0LL
58. 1 1.0 2 1.0
59. LOAD COMB 202 1.0DL+0.75LL
60. 1 1.0 2 0.75
61. LOAD COMB 203 1.0DL+1.0WLX1
62. 1 1.0 3 1.0
63. LOAD COMB 204 1.0DL+1.0WLX2
64. 1 1.0 4 1.0
65. LOAD COMB 205 1.0DL+1.0WLZ1
66. 1 1.0 5 1.0
67. LOAD COMB 206 1.0DL+1.0WLZ2
68. 1 1.0 6 1.0
69. LOAD COMB 207 1.0DL+0.75LL+0.75WLX1
70. 1 1.0 2 0.75 3 0.75
71. LOAD COMB 208 1.0DL+0.75LL+0.75WLX2
72. 1 1.0 2 0.75 4 0.75
73. LOAD COMB 209 1.0DL+0.75LL+0.75WLZ1
74. 1 1.0 2 0.75 5 0.75
75. LOAD COMB 210 1.0DL+0.75LL+0.75WLZ2
76. 1 1.0 2 0.75 6 0.75
77. LOAD COMB 211 0.6DL+1.0WLX1
78. 1 0.6 3 1.0
79. LOAD COMB 212 0.6DL+1.0WLX2
80. 1 0.6 4 1.0
81. LOAD COMB 213 0.6DL+1.0WLZ1
82. 1 0.6 5 1.0
83. LOAD COMB 214 0.6DL+1.0WLZ2
84. 1 0.6 6 1.0
85. ********
86. ********
87. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112065.3 MB

88. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 8,2020 TIME= 11:36:46 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE FOR F2 DESIGN

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  8, 2020                *
*           Time=    11:38:50                    *
*                                                *
*      USER ID: LT Construction                 *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 2.684 0 0; 3 0 0 4.9; 4 2.684 0 4.9; 5 3.342 0 0; 6 3.342 0 4.9
    10. 7 -0.658 0 0; 8 -0.658 0 4.9; 9 0 0 -0.55; 10 2.684 0 -0.55; 11 3.342 0 -0.55
    11. 12 -0.658 0 -0.55; 13 0 0 5.45; 14 2.684 0 5.45; 15 3.342 0 5.45
    12. 16 -0.658 0 5.45; 17 0 2.8 0; 18 2.684 2.8 0; 19 0 2.8 4.9; 20 2.684 2.8 4.9
    13. 21 -0.229 0 0; 22 -0.229 0 -0.25; 23 -0.658 0 -0.25; 24 0 0 -0.25
    14. 25 -0.229 0 -0.55; 26 2.913 0 0; 27 2.913 0 -0.25; 28 2.684 0 -0.25
    15. 29 3.342 0 -0.25; 30 2.913 0 -0.55; 31 -0.229 0 5.45; 32 -0.229 0 5.15
    16. 33 -0.658 0 5.15; 34 0 0 5.15; 35 -0.229 0 4.9; 36 2.913 0 5.45
    17. 37 2.913 0 5.15; 38 2.684 0 5.15; 39 3.342 0 5.15; 40 2.913 0 4.9
    18. 41 0.244 0 0; 42 0.244 0 -0.25; 43 0.488 0 0; 44 0.488 0 -0.25; 45 0.732 0 0
    19. 46 0.732 0 -0.25; 47 0.976 0 0; 48 0.976 0 -0.25; 49 1.22 0 0; 50 1.22 0 -0.25
    20. 51 1.464 0 0; 52 1.464 0 -0.25; 53 1.708 0 0; 54 1.708 0 -0.25; 55 1.952 0 0
    21. 56 1.952 0 -0.25; 57 2.196 0 0; 58 2.196 0 -0.25; 59 2.44 0 0; 60 2.44 0 -0.25
    22. 61 0.244 0 -0.55; 62 0.488 0 -0.55; 63 0.732 0 -0.55; 64 0.976 0 -0.55
    23. 65 1.22 0 -0.55; 66 1.464 0 -0.55; 67 1.708 0 -0.55; 68 1.952 0 -0.55
    24. 69 2.196 0 -0.55; 70 2.44 0 -0.55; 71 -0.229 0 4.655; 72 -0.658 0 4.655
    25. 73 0 0 4.655; 74 -0.229 0 4.41; 75 -0.658 0 4.41; 76 0 0 4.41
    26. 77 -0.229 0 4.165; 78 -0.658 0 4.165; 79 0 0 4.165; 80 -0.229 0 3.92
    27. 81 -0.658 0 3.92; 82 0 0 3.92; 83 -0.229 0 3.675; 84 -0.658 0 3.675
    28. 85 0 0 3.675; 86 -0.229 0 3.43; 87 -0.658 0 3.43; 88 0 0 3.43
    29. 89 -0.229 0 3.185; 90 -0.658 0 3.185; 91 0 0 3.185; 92 -0.229 0 2.94
    30. 93 -0.658 0 2.94; 94 0 0 2.94; 95 -0.229 0 2.695; 96 -0.658 0 2.695
    31. 97 0 0 2.695; 98 -0.229 0 2.45; 99 -0.658 0 2.45; 100 0 0 2.45
    32. 101 -0.229 0 2.205; 102 -0.658 0 2.205; 103 0 0 2.205; 104 -0.229 0 1.96
    33. 105 -0.658 0 1.96; 106 0 0 1.96; 107 -0.229 0 1.715; 108 -0.658 0 1.715
    34. 109 0 0 1.715; 110 -0.229 0 1.47; 111 -0.658 0 1.47; 112 0 0 1.47
    35. 113 -0.229 0 1.225; 114 -0.658 0 1.225; 115 0 0 1.225; 116 -0.229 0 0.979999
    36. 117 -0.658 0 0.979999; 118 0 0 0.979999; 119 -0.229 0 0.734999
    37. 120 -0.658 0 0.734999; 121 0 0 0.734999; 122 -0.229 0 0.489999
    38. 123 -0.658 0 0.489999; 124 0 0 0.489999; 125 -0.229 0 0.244999
```

39. 126 -0.658 0 0.244999; 127 0 0 0.244999; 128 2.913 0 4.655; 129 2.684 0 4.655
40. 130 3.342 0 4.655; 131 2.913 0 4.41; 132 2.684 0 4.41; 133 3.342 0 4.41
41. 134 2.913 0 4.165; 135 2.684 0 4.165; 136 3.342 0 4.165; 137 2.913 0 3.92
42. 138 2.684 0 3.92; 139 3.342 0 3.92; 140 2.913 0 3.675; 141 2.684 0 3.675
43. 142 3.342 0 3.675; 143 2.913 0 3.43; 144 2.684 0 3.43; 145 3.342 0 3.43
44. 146 2.913 0 3.185; 147 2.684 0 3.185; 148 3.342 0 3.185; 149 2.913 0 2.94
45. 150 2.684 0 2.94; 151 3.342 0 2.94; 152 2.913 0 2.695; 153 2.684 0 2.695
46. 154 3.342 0 2.695; 155 2.913 0 2.45; 156 2.684 0 2.45; 157 3.342 0 2.45
47. 158 2.913 0 2.205; 159 2.684 0 2.205; 160 3.342 0 2.205; 161 2.913 0 1.96
48. 162 2.684 0 1.96; 163 3.342 0 1.96; 164 2.913 0 1.715; 165 2.684 0 1.715
49. 166 3.342 0 1.715; 167 2.913 0 1.47; 168 2.684 0 1.47; 169 3.342 0 1.47
50. 170 2.913 0 1.225; 171 2.684 0 1.225; 172 3.342 0 1.225; 173 2.913 0 0.979999
51. 174 2.684 0 0.979999; 175 3.342 0 0.979999; 176 2.913 0 0.734999
52. 177 2.684 0 0.734999; 178 3.342 0 0.734999; 179 2.913 0 0.489999
53. 180 2.684 0 0.489999; 181 3.342 0 0.489999; 182 2.913 0 0.244999
54. 183 2.684 0 0.244999; 184 3.342 0 0.244999; 185 0.244 0 5.45; 186 0.244 0 5.15
55. 187 0.488 0 5.45; 188 0.488 0 5.15; 189 0.732 0 5.45; 190 0.732 0 5.15
56. 191 0.976 0 5.45; 192 0.976 0 5.15; 193 1.22 0 5.45; 194 1.22 0 5.15
57. 195 1.464 0 5.45; 196 1.464 0 5.15; 197 1.708 0 5.45; 198 1.708 0 5.15
58. 199 1.952 0 5.45; 200 1.952 0 5.15; 201 2.196 0 5.45; 202 2.196 0 5.15
59. 203 2.44 0 5.45; 204 2.44 0 5.15; 205 0.244 0 4.9; 206 0.488 0 4.9
60. 207 0.732 0 4.9; 208 0.976 0 4.9; 209 1.22 0 4.9; 210 1.464 0 4.9
61. 211 1.708 0 4.9; 212 1.952 0 4.9; 213 2.196 0 4.9; 214 2.44 0 4.9
62. 215 0.244 0 4.655; 216 0.488 0 4.655; 217 0.732 0 4.655; 218 0.976 0 4.655
63. 219 1.22 0 4.655; 220 1.464 0 4.655; 221 1.708 0 4.655; 222 1.952 0 4.655
64. 223 2.196 0 4.655; 224 2.44 0 4.655; 225 0.244 0 4.41; 226 0.488 0 4.41
65. 227 0.732 0 4.41; 228 0.976 0 4.41; 229 1.22 0 4.41; 230 1.464 0 4.41
66. 231 1.708 0 4.41; 232 1.952 0 4.41; 233 2.196 0 4.41; 234 2.44 0 4.41
67. 235 0.244 0 4.165; 236 0.488 0 4.165; 237 0.732 0 4.165; 238 0.976 0 4.165
68. 239 1.22 0 4.165; 240 1.464 0 4.165; 241 1.708 0 4.165; 242 1.952 0 4.165
69. 243 2.196 0 4.165; 244 2.44 0 4.165; 245 0.244 0 3.92; 246 0.488 0 3.92
70. 247 0.732 0 3.92; 248 0.976 0 3.92; 249 1.22 0 3.92; 250 1.464 0 3.92
71. 251 1.708 0 3.92; 252 1.952 0 3.92; 253 2.196 0 3.92; 254 2.44 0 3.92
72. 255 0.244 0 3.675; 256 0.488 0 3.675; 257 0.732 0 3.675; 258 0.976 0 3.675
73. 259 1.22 0 3.675; 260 1.464 0 3.675; 261 1.708 0 3.675; 262 1.952 0 3.675
74. 263 2.196 0 3.675; 264 2.44 0 3.675; 265 0.244 0 3.43; 266 0.488 0 3.43
75. 267 0.732 0 3.43; 268 0.976 0 3.43; 269 1.22 0 3.43; 270 1.464 0 3.43
76. 271 1.708 0 3.43; 272 1.952 0 3.43; 273 2.196 0 3.43; 274 2.44 0 3.43
77. 275 0.244 0 3.185; 276 0.488 0 3.185; 277 0.732 0 3.185; 278 0.976 0 3.185
78. 279 1.22 0 3.185; 280 1.464 0 3.185; 281 1.708 0 3.185; 282 1.952 0 3.185
79. 283 2.196 0 3.185; 284 2.44 0 3.185; 285 0.244 0 2.94; 286 0.488 0 2.94
80. 287 0.732 0 2.94; 288 0.976 0 2.94; 289 1.22 0 2.94; 290 1.464 0 2.94
81. 291 1.708 0 2.94; 292 1.952 0 2.94; 293 2.196 0 2.94; 294 2.44 0 2.94
82. 295 0.244 0 2.695; 296 0.488 0 2.695; 297 0.732 0 2.695; 298 0.976 0 2.695
83. 299 1.22 0 2.695; 300 1.464 0 2.695; 301 1.708 0 2.695; 302 1.952 0 2.695
84. 303 2.196 0 2.695; 304 2.44 0 2.695; 305 0.244 0 2.45; 306 0.488 0 2.45
85. 307 0.732 0 2.45; 308 0.976 0 2.45; 309 1.22 0 2.45; 310 1.464 0 2.45
86. 311 1.708 0 2.45; 312 1.952 0 2.45; 313 2.196 0 2.45; 314 2.44 0 2.45
87. 315 0.244 0 2.205; 316 0.488 0 2.205; 317 0.732 0 2.205; 318 0.976 0 2.205
88. 319 1.22 0 2.205; 320 1.464 0 2.205; 321 1.708 0 2.205; 322 1.952 0 2.205
89. 323 2.196 0 2.205; 324 2.44 0 2.205; 325 0.244 0 1.96; 326 0.488 0 1.96
90. 327 0.732 0 1.96; 328 0.976 0 1.96; 329 1.22 0 1.96; 330 1.464 0 1.96
91. 331 1.708 0 1.96; 332 1.952 0 1.96; 333 2.196 0 1.96; 334 2.44 0 1.96
92. 335 0.244 0 1.715; 336 0.488 0 1.715; 337 0.732 0 1.715; 338 0.976 0 1.715
93. 339 1.22 0 1.715; 340 1.464 0 1.715; 341 1.708 0 1.715; 342 1.952 0 1.715
94. 343 2.196 0 1.715; 344 2.44 0 1.715; 345 0.244 0 1.47; 346 0.488 0 1.47

```
 95. 347 0.732 0 1.47; 348 0.976 0 1.47; 349 1.22 0 1.47; 350 1.464 0 1.47
 96. 351 1.708 0 1.47; 352 1.952 0 1.47; 353 2.196 0 1.47; 354 2.44 0 1.47
 97. 355 0.244 0 1.225; 356 0.488 0 1.225; 357 0.732 0 1.225; 358 0.976 0 1.225
 98. 359 1.22 0 1.225; 360 1.464 0 1.225; 361 1.708 0 1.225; 362 1.952 0 1.225
 99. 363 2.196 0 1.225; 364 2.44 0 1.225; 365 0.244 0 0.979999
100. 366 0.488 0 0.979999; 367 0.732 0 0.979999; 368 0.976 0 0.979999
101. 369 1.22 0 0.979999; 370 1.464 0 0.979999; 371 1.708 0 0.979999
102. 372 1.952 0 0.979999; 373 2.196 0 0.979999; 374 2.44 0 0.979999
103. 375 0.244 0 0.734999; 376 0.488 0 0.734999; 377 0.732 0 0.734999
104. 378 0.976 0 0.734999; 379 1.22 0 0.734999; 380 1.464 0 0.734999
105. 381 1.708 0 0.734999; 382 1.952 0 0.734999; 383 2.196 0 0.734999
106. 384 2.44 0 0.734999; 385 0.244 0 0.489999; 386 0.488 0 0.489999
107. 387 0.732 0 0.489999; 388 0.976 0 0.489999; 389 1.22 0 0.489999
108. 390 1.464 0 0.489999; 391 1.708 0 0.489999; 392 1.952 0 0.489999
109. 393 2.196 0 0.489999; 394 2.44 0 0.489999; 395 0.244 0 0.244999
110. 396 0.488 0 0.244999; 397 0.732 0 0.244999; 398 0.976 0 0.244999
111. 399 1.22 0 0.244999; 400 1.464 0 0.244999; 401 1.708 0 0.244999
112. 402 1.952 0 0.244999; 403 2.196 0 0.244999; 404 2.44 0 0.244999
113. 405 -0.4435 0 0; 406 -0.4435 0 -0.25; 407 -0.4435 0 -0.55; 408 3.1275 0 0
114. 409 3.1275 0 -0.25; 410 3.1275 0 -0.55; 411 -0.4435 0 5.45; 412 -0.4435 0 5.15
115. 413 -0.4435 0 4.9; 414 3.1275 0 5.45; 415 3.1275 0 5.15; 416 3.1275 0 4.9
116. 417 -0.4435 0 4.655; 418 -0.4435 0 4.41; 419 -0.4435 0 4.165
117. 420 -0.4435 0 3.92; 421 -0.4435 0 3.675; 422 -0.4435 0 3.43
118. 423 -0.4435 0 3.185; 424 -0.4435 0 2.94; 425 -0.4435 0 2.695
119. 426 -0.4435 0 2.45; 427 -0.4435 0 2.205; 428 -0.4435 0 1.96
120. 429 -0.4435 0 1.715; 430 -0.4435 0 1.47; 431 -0.4435 0 1.225
121. 432 -0.4435 0 0.979999; 433 -0.4435 0 0.734999; 434 -0.4435 0 0.489999
122. 435 -0.4435 0 0.244999; 436 3.1275 0 4.655; 437 3.1275 0 4.41
123. 438 3.1275 0 4.165; 439 3.1275 0 3.92; 440 3.1275 0 3.675; 441 3.1275 0 3.43
124. 442 3.1275 0 3.185; 443 3.1275 0 2.94; 444 3.1275 0 2.695; 445 3.1275 0 2.45
125. 446 3.1275 0 2.205; 447 3.1275 0 1.96; 448 3.1275 0 1.715; 449 3.1275 0 1.47
126. 450 3.1275 0 1.225; 451 3.1275 0 0.979999; 452 3.1275 0 0.734999
127. 453 3.1275 0 0.489999; 454 3.1275 0 0.244999; 455 0 0.2 0; 456 2.684 0.2 0
128. 457 0 0.2 4.9; 458 2.684 0.2 4.9
129. MEMBER INCIDENCES
130. 25 1 455; 26 2 456; 27 3 457; 28 4 458; 626 455 17; 627 456 18; 628 457 19
131. 629 458 20
132. ELEMENT INCIDENCES SHELL
133. 42 21 1 24 22; 45 22 24 9 25; 48 2 26 27 28; 52 28 27 30 10; 58 31 13 34 32
134. 61 32 34 3 35; 64 14 36 37 38; 68 38 37 40 4; 71 1 41 42 24; 73 41 43 44 42
135. 75 43 45 46 44; 77 45 47 48 46; 79 47 49 50 48; 81 49 51 52 50; 83 51 53 54 52
136. 85 53 55 56 54; 87 55 57 58 56; 89 57 59 60 58; 90 59 2 28 60; 92 24 42 61 9
137. 94 42 44 62 61; 96 44 46 63 62; 98 46 48 64 63; 100 48 50 65 64
138. 102 50 52 66 65; 104 52 54 67 66; 106 54 56 68 67; 108 56 58 69 68
139. 110 58 60 70 69; 111 60 28 10 70; 115 35 3 73 71; 119 71 73 76 74
140. 123 74 76 79 77; 127 77 79 82 80; 131 80 82 85 83; 135 83 85 88 86
141. 139 86 88 91 89; 143 89 91 94 92; 147 92 94 97 95; 151 95 97 100 98
142. 155 98 100 103 101; 159 101 103 106 104; 163 104 106 109 107
143. 167 107 109 112 110; 171 110 112 115 113; 175 113 115 118 116
144. 179 116 118 121 119; 183 119 121 124 122; 187 122 124 127 125
145. 189 125 127 1 21; 191 4 40 128 129; 195 129 128 131 132; 199 132 131 134 135
146. 203 135 134 137 138; 207 138 137 140 141; 211 141 140 143 144
147. 215 144 143 146 147; 219 147 146 149 150; 223 150 149 152 153
148. 227 153 152 155 156; 231 156 155 158 159; 235 159 158 161 162
149. 239 162 161 164 165; 243 165 164 167 168; 247 168 167 170 171
150. 251 171 170 173 174; 255 174 173 176 177; 259 177 176 179 180
```

151. 263 180 179 182 183; 266 183 182 26 2; 269 13 185 186 34; 271 185 187 188 186
152. 273 187 189 190 188; 275 189 191 192 190; 277 191 193 194 192
153. 279 193 195 196 194; 281 195 197 198 196; 283 197 199 200 198
154. 285 199 201 202 200; 287 201 203 204 202; 288 203 14 38 204; 290 34 186 205 3
155. 292 186 188 206 205; 294 188 190 207 206; 296 190 192 208 207
156. 298 192 194 209 208; 300 194 196 210 209; 302 196 198 211 210
157. 304 198 200 212 211; 306 200 202 213 212; 308 202 204 214 213
158. 309 204 38 4 214; 310 3 205 215 73; 311 205 206 216 215; 312 206 207 217 216
159. 313 207 208 218 217; 314 208 209 219 218; 315 209 210 220 219
160. 316 210 211 221 220; 317 211 212 222 221; 318 212 213 223 222
161. 319 213 214 224 223; 320 214 4 129 224; 321 73 215 225 76; 322 215 216 226 225
162. 323 216 217 227 226; 324 217 218 228 227; 325 218 219 229 228
163. 326 219 220 230 229; 327 220 221 231 230; 328 221 222 232 231
164. 329 222 223 233 232; 330 223 224 234 233; 331 224 129 132 234
165. 332 76 225 235 79; 333 225 226 236 235; 334 226 227 237 236
166. 335 227 228 238 237; 336 228 229 239 238; 337 229 230 240 239
167. 338 230 231 241 240; 339 231 232 242 241; 340 232 233 243 242
168. 341 233 234 244 243; 342 234 132 135 244; 343 79 235 245 82
169. 344 235 236 246 245; 345 236 237 247 246; 346 237 238 248 247
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172. 353 244 135 138 254; 354 82 245 255 85; 355 245 246 256 255
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175. 362 252 253 263 262; 363 253 254 264 263; 364 254 138 141 264
176. 365 85 255 265 88; 366 255 256 266 265; 367 256 257 267 266
177. 368 257 258 268 267; 369 258 259 269 268; 370 259 260 270 269
178. 371 260 261 271 270; 372 261 262 272 271; 373 262 263 273 272
179. 374 263 264 274 273; 375 264 141 144 274; 376 88 265 275 91
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185. 392 279 280 290 289; 393 280 281 291 290; 394 281 282 292 291
186. 395 282 283 293 292; 396 283 284 294 293; 397 284 147 150 294
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188. 401 287 288 298 297; 402 288 289 299 298; 403 289 290 300 299
189. 404 290 291 301 300; 405 291 292 302 301; 406 292 293 303 302
190. 407 293 294 304 303; 408 294 150 153 304; 409 97 295 305 100
191. 410 295 296 306 305; 411 296 297 307 306; 412 297 298 308 307
192. 413 298 299 309 308; 414 299 300 310 309; 415 300 301 311 310
193. 416 301 302 312 311; 417 302 303 313 312; 418 303 304 314 313
194. 419 304 153 156 314; 420 100 305 315 103; 421 305 306 316 315
195. 422 306 307 317 316; 423 307 308 318 317; 424 308 309 319 318
196. 425 309 310 320 319; 426 310 311 321 320; 427 311 312 322 321
197. 428 312 313 323 322; 429 313 314 324 323; 430 314 156 159 324
198. 431 103 315 325 106; 432 315 316 326 325; 433 316 317 327 326
199. 434 317 318 328 327; 435 318 319 329 328; 436 319 320 330 329
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201. 440 323 324 334 333; 441 324 159 162 334; 442 106 325 335 109
202. 443 325 326 336 335; 444 326 327 337 336; 445 327 328 338 337
203. 446 328 329 339 338; 447 329 330 340 339; 448 330 331 341 340
204. 449 331 332 342 341; 450 332 333 343 342; 451 333 334 344 343
205. 452 334 162 165 344; 453 109 335 345 112; 454 335 336 346 345
206. 455 336 337 347 346; 456 337 338 348 347; 457 338 339 349 348

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235. 550 75 418 419 78; 551 418 74 77 419; 552 78 419 420 81; 553 419 77 80 420
236. 554 81 420 421 84; 555 420 80 83 421; 556 84 421 422 87; 557 421 83 86 422
237. 558 87 422 423 90; 559 422 86 89 423; 560 90 423 424 93; 561 423 89 92 424
238. 562 93 424 425 96; 563 424 92 95 425; 564 96 425 426 99; 565 425 95 98 426
239. 566 99 426 427 102; 567 426 98 101 427; 568 102 427 428 105
240. 569 427 101 104 428; 570 105 428 429 108; 571 428 104 107 429
241. 572 108 429 430 111; 573 429 107 110 430; 574 111 430 431 114
242. 575 430 110 113 431; 576 114 431 432 117; 577 431 113 116 432
243. 578 117 432 433 120; 579 432 116 119 433; 580 120 433 434 123
244. 581 433 119 122 434; 582 123 434 435 126; 583 434 122 125 435
245. 584 126 435 405 7; 585 435 125 21 405; 586 40 416 436 128; 587 416 6 130 436
246. 588 128 436 437 131; 589 436 130 133 437; 590 131 437 438 134
247. 591 437 133 136 438; 592 134 438 439 137; 593 438 136 139 439
248. 594 137 439 440 140; 595 439 139 142 440; 596 140 440 441 143
249. 597 440 142 145 441; 598 143 441 442 146; 599 441 145 148 442
250. 600 146 442 443 149; 601 442 148 151 443; 602 149 443 444 152
251. 603 443 151 154 444; 604 152 444 445 155; 605 444 154 157 445
252. 606 155 445 446 158; 607 445 157 160 446; 608 158 446 447 161
253. 609 446 160 163 447; 610 161 447 448 164; 611 447 163 166 448
254. 612 164 448 449 167; 613 448 166 169 449; 614 167 449 450 170
255. 615 449 169 172 450; 616 170 450 451 173; 617 450 172 175 451
256. 618 173 451 452 176; 619 451 175 178 452; 620 176 452 453 179
257. 621 452 178 181 453; 622 179 453 454 182; 623 453 181 184 454
258. 624 182 454 408 26; 625 454 184 5 408
259. START GROUP DEFINITION
260. ELEMENT
261. _RAFT 42 45 48 52 58 61 64 68 71 73 75 77 79 81 83 85 87 89 90 92 94 96 98 -
262. 100 102 104 106 108 110 111 115 119 123 127 131 135 139 143 147 151 155 159 -
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263. 163 167 171 175 179 183 187 189 191 195 199 203 207 211 215 219 223 227 231 -
264. 235 239 243 247 251 255 259 263 266 269 271 273 275 277 279 281 283 285 287 -
265. 288 290 292 294 296 298 300 302 304 306 308 TO 625
266. _RAFT_MOM 45 52 58 64 73 75 77 79 81 83 85 87 89 92 94 96 98 100 102 104 106 -
267. 108 110 111 119 123 127 131 135 139 143 147 151 155 159 163 167 171 175 179 -
268. 183 187 195 199 203 207 211 215 219 223 227 231 235 239 243 247 251 255 259 -
269. 263 269 271 273 275 277 279 281 283 285 287 288 292 294 296 298 300 302 304 -
270. 306 308 311 TO 319 321 TO 518 520 TO 528 530 TO 625
271. _PLATE_SHEAR 75 77 79 81 83 85 87 96 98 100 102 104 106 108 123 127 131 135 -
272. 139 143 147 151 155 159 163 167 171 175 179 183 199 203 207 211 215 219 223 -
273. 227 231 235 239 243 247 251 255 259 273 275 277 279 281 283 285 294 296 298 -
274. 300 302 304 306 312 TO 318 323 TO 329 332 TO 507 510 TO 516 521 TO 527 530 -
275. 532 535 537 538 540 543 545 546 548 550 TO 582 584 587 589 TO 621 623 625
276. MEMBER
277. _C1 626 TO 629
278. END GROUP DEFINITION
279. ELEMENT PROPERTY
280. 42 45 48 52 58 61 64 68 71 73 75 77 79 81 83 85 87 89 90 92 94 96 98 100 102 -
281. 104 106 108 110 111 115 119 123 127 131 135 139 143 147 151 155 159 163 167 -
282. 171 175 179 183 187 189 191 195 199 203 207 211 215 219 223 227 231 235 239 -
283. 243 247 251 255 259 263 266 269 271 273 275 277 279 281 283 285 287 288 290 -
284. 292 294 296 298 300 302 304 306 308 TO 625 THICKNESS 0.4
285. DEFINE MATERIAL START
286. ISOTROPIC CONCRETE
287. E 2.17185E+007
288. POISSON 0.17
289. DENSITY 25
290. ALPHA 1E-005
291. DAMP 0.05
292. TYPE CONCRETE
293. STRENGTH FCU 27579
294. END DEFINE MATERIAL
295. MEMBER PROPERTY AMERICAN
296. 25 TO 28 626 TO 629 PRIS YD 0.5 ZD 0.5
297. CONSTANTS
298. MATERIAL CONCRETE ALL
299. **********************
300. ************************************************************
301. SUPPORTS
302. 42 45 48 52 58 61 64 68 71 73 75 77 79 81 83 85 87 89 90 92 94 96 98 100 102 -
303. 104 106 108 110 111 115 119 123 127 131 135 139 143 147 151 155 159 163 167 -
304. 171 175 179 183 187 189 191 195 199 203 207 211 215 219 223 227 231 235 239 -
305. 243 247 251 255 259 263 266 269 271 273 275 277 279 281 283 285 287 288 290 -
306. 292 294 296 298 300 302 304 306 308 TO 624 -
307. 625 PLATE MAT DIRECT Y SUBGRADE 60000 PRINT
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JOINT	INFLUENCE AREA
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2	0.058534
3	0.058534
4	0.058534
5	0.026544
6	0.026544
7	0.026544

8	0.026544
9	0.035475
10	0.035475
11	0.016087
12	0.016087
13	0.035475
14	0.035475
15	0.016087
16	0.016087
21	0.054883
22	0.060981
23	0.029494
24	0.065038
25	0.033263
26	0.054883
27	0.060981
28	0.065038
29	0.029494
30	0.033263
31	0.033262
32	0.060981
33	0.029494
34	0.065037
35	0.054883
36	0.033262
37	0.060981
38	0.065037
39	0.029494
40	0.054883
41	0.060390
42	0.067100
43	0.060390
44	0.067100
45	0.060390
46	0.067100
47	0.060390
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67	0.036600

68	0.036600
69	0.036600
70	0.036600
71	0.054329
72	0.026276
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74	0.054329
75	0.026276
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77	0.054329
78	0.026276
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83	0.054329
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85	0.057942
86	0.054329
87	0.026276
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452	0.052552
453	0.052552
454	0.052552
TOTAL	24.000046

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308. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
309. JOINT LOAD
310. 17 FY -40
311. 18 FY -20
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312. 19 FY -50 FZ 10
313. 20 FY -30
314. SELFWEIGHT Y -1 LIST ALL
315. ***
316. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
317. JOINT LOAD
318. 17 FY -110 FZ -10
319. 18 FY -40
320. 19 FY -140 FZ 30
321. 20 FY -80
322. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
323. JOINT LOAD
324. 17 FX 20 FY 90
325. 18 FX 20 FY -90
326. 19 FX 30 FY 110
327. 20 FX 30 FY -120
328. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
329. JOINT LOAD
330. 17 FX -20 FY -90
331. 18 FX -20 FY 90
332. 19 FX -30 FY -110
333. 20 FX -30 FY 120
334. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
335. JOINT LOAD
336. 17 FY 30 FZ 30
337. 18 FY 30 FZ 20
338. 19 FY -40 FZ 10
339. 20 FY -30
340. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
341. JOINT LOAD
342. 17 FY -30 FZ -30
343. 18 FY -30 FZ -20
344. 19 FY 40 FZ -10
345. 20 FY 30
346. LOAD 7 LOADTYPE SOIL  TITLE SOIL LOAD (HL)
347. ELEMENT LOAD
348. 42 45 48 52 58 61 64 68 71 73 75 77 79 81 83 85 87 89 90 92 94 96 98 100 102 -
349. 104 106 108 110 111 115 119 123 127 131 135 139 143 147 151 155 159 163 167 -
350. 171 175 179 183 187 189 191 195 199 203 207 211 215 219 223 227 231 235 239 -
351. 243 247 251 255 259 263 266 269 271 273 275 277 279 281 283 285 287 288 290 -
352. 292 294 296 298 300 302 304 306 308 TO 625 PR -19.8
353. **************************************************
354. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
355. LOAD 101 1.4DL
356. REPEAT LOAD
357. 1 1.4
358. LOAD 102 1.4DL+1.7LL
359. REPEAT LOAD
360. 1 1.4 2 1.7
361. LOAD 103 1.2DL+1.0LL+1.6WLX1
362. REPEAT LOAD
363. 1 1.2 2 1.0 3 1.6
364. LOAD 104 1.2DL+1.0LL+1.6WLX2
365. REPEAT LOAD
366. 1 1.2 2 1.0 4 1.6
367. LOAD 105 1.2DL+1.0LL+1.6WLZ1
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368. REPEAT LOAD
369. 1 1.2 2 1.0 5 1.6
370. LOAD 106 1.2DL+1.0LL+1.6WLZ2
371. REPEAT LOAD
372. 1 1.2 2 1.0 6 1.6
373. LOAD 107 0.9DL+1.6HL+1.6WLX1
374. REPEAT LOAD
375. 1 0.9 7 1.6 3 1.6
376. LOAD 108 0.9DL+1.6HL+1.6WLX2
377. REPEAT LOAD
378. 1 0.9 7 1.6 4 1.6
379. LOAD 109 0.9DL+1.6HL+1.6WLZ1
380. REPEAT LOAD
381. 1 0.9 7 1.6 5 1.6
382. LOAD 110 0.9DL+1.6HL+1.6WLZ2
383. REPEAT LOAD
384. 1 0.9 7 1.6 6 1.6
385. ***************
386. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 458 NUMBER OF MEMBERS 8
NUMBER OF PLATES 408 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 450

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 454/ 30/ 93 DOF
TOTAL PRIMARY LOAD CASES = 17, TOTAL DEGREES OF FREEDOM = 1398
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 131 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.8/ 112065.1 MB

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387. CHANGE
388. SUPPORTS
389. 42 45 48 52 58 61 64 68 71 73 75 77 79 81 83 85 87 89 90 92 94 96 98 100 102 -
390. 104 106 108 110 111 115 119 123 127 131 135 139 143 147 151 155 159 163 167 -
391. 171 175 179 183 187 189 191 195 199 203 207 211 215 219 223 227 231 235 239 -
392. 243 247 251 255 259 263 266 269 271 273 275 277 279 281 283 285 287 288 290 -
393. 292 294 296 298 300 302 304 306 308 TO 624 -
394. 625 PLATE MAT DIRECT Y SUBGRADE 60000 PRINT COMPRESSION
```

JOINT	INFLUENCE AREA
1	0.058534

2	0.058534
3	0.058534
4	0.058534
5	0.026544
6	0.026544
7	0.026544
8	0.026544
9	0.035475
10	0.035475
11	0.016087
12	0.016087
13	0.035475
14	0.035475
15	0.016087
16	0.016087
21	0.054883
22	0.060981
23	0.029494
24	0.065038
25	0.033263
26	0.054883
27	0.060981
28	0.065038
29	0.029494
30	0.033263
31	0.033262
32	0.060981
33	0.029494
34	0.065037
35	0.054883
36	0.033262
37	0.060981
38	0.065037
39	0.029494
40	0.054883
41	0.060390
42	0.067100
43	0.060390
44	0.067100
45	0.060390
46	0.067100
47	0.060390
48	0.067100
49	0.060390
50	0.067100
51	0.060390
52	0.067100
53	0.060390
54	0.067100
55	0.060390
56	0.067100
57	0.060390
58	0.067100
59	0.060390
60	0.067100
61	0.036600

62	0.036600
63	0.036600
64	0.036600
65	0.036600
66	0.036600
67	0.036600
68	0.036600
69	0.036600
70	0.036600
71	0.054329
72	0.026276
73	0.057942
74	0.054329
75	0.026276
76	0.057942
77	0.054329
78	0.026276
79	0.057943
80	0.054329
81	0.026276
82	0.057942
83	0.054329
84	0.026276
85	0.057942
86	0.054329
87	0.026276
88	0.057943
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90	0.026276
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96	0.026276
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99	0.026276
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102	0.026276
103	0.057943
104	0.054329
105	0.026276
106	0.057943
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108	0.026276
109	0.057942
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111	0.026276
112	0.057942
113	0.054329
114	0.026276
115	0.057943
116	0.054329
117	0.026276

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143	0.054329
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399	0.059780
400	0.059780
401	0.059780
402	0.059780
403	0.059780
404	0.059780
405	0.053089
406	0.058987
407	0.032175
408	0.053089
409	0.058987
410	0.032175
411	0.032175
412	0.058987
413	0.053089
414	0.032175
415	0.058987
416	0.053089
417	0.052552
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419	0.052553
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422	0.052553
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425	0.052552
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429	0.052552
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431	0.052553
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433	0.052552
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448	0.052552
449	0.052552
450	0.052553
451	0.052553
452	0.052552
453	0.052552

```
       454          0.052552
    TOTAL          24.000046
395. ********************************************************************************
396. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
397. ************************************************************
398. LOAD 201 1.0DL+1.0HL+1.0LL
399. REPEAT LOAD
400. 1 1.0 7 1.0 2 1.0
401. LOAD 202 1.0DL+1.0HL+0.75LL
402. REPEAT LOAD
403. 1 1.0 7 1.0 2 0.75
404. LOAD 203 1.0DL+1.0HL+1.0WLX1
405. REPEAT LOAD
406. 1 1.0 7 1.0 3 1.0
407. LOAD 204 1.0DL+1.0HL+1.0WLX2
408. REPEAT LOAD
409. 1 1.0 7 1.0 4 1.0
410. LOAD 205 1.0DL+1.0HL+1.0WLZ1
411. REPEAT LOAD
412. 1 1.0 7 1.0 5 1.0
413. LOAD 206 1.0DL+1.0HL+1.0WLZ2
414. REPEAT LOAD
415. 1 1.0 7 1.0 6 1.0
416. LOAD 207 1.0DL+1.0HL+0.75LL+0.75WLX1
417. REPEAT LOAD
418. 1 1.0 7 1.0 2 0.75 3 0.75
419. LOAD 208 1.0DL+1.0HL+0.75LL+0.75WLX2
420. REPEAT LOAD
421. 1 1.0 7 1.0 2 0.75 4 0.75
422. LOAD 209 1.0DL+1.0HL+0.75LL+0.75WLZ1
423. REPEAT LOAD
424. 1 1.0 7 1.0 2 0.75 5 0.75
425. LOAD 210 1.0DL+1.0HL+0.75LL+0.75WLZ2
426. REPEAT LOAD
427. 1 1.0 7 1.0 2 0.75 6 0.75
428. LOAD 211 0.6DL+1.0HL+1.0WLX1
429. REPEAT LOAD
430. 1 0.6 7 1.0 3 1.0
431. LOAD 212 0.6DL+1.0HL+1.0WLX2
432. REPEAT LOAD
433. 1 0.6 7 1.0 4 1.0
434. LOAD 213 0.6DL+1.0HL+1.0WLZ1
435. REPEAT LOAD
436. 1 0.6 7 1.0 5 1.0
437. LOAD 214 0.6DL+1.0HL+1.0WLZ2
438. REPEAT LOAD
439. 1 0.6 7 1.0 6 1.0
440. ********
441. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 203

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 211

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 212

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

442. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 8,2020 TIME= 11:39:31 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 4 STAAD INPUT & OUTPUT FILE FOR F2 STABILITY

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  8, 2020                *
*           Time=    11:43:17                    *
*                                                *
*      USER ID: LT Construction                 *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 2.684 0 0; 3 0 0 4.9; 4 2.684 0 4.9; 17 0 2.8 0; 18 2.684 2.8 0
     9. 19 0 2.8 4.9; 20 2.684 2.8 4.9
    10. MEMBER INCIDENCES
    11. 25 1 17; 26 2 18; 27 3 19; 28 4 20
    12. DEFINE MATERIAL START
    13. ISOTROPIC CONCRETE
    14. E 2.17185E+007
    15. POISSON 0.17
    16. DENSITY 25
    17. ALPHA 1E-005
    18. DAMP 0.05
    19. TYPE CONCRETE
    20. STRENGTH FCU 27579
    21. END DEFINE MATERIAL
    22. MEMBER PROPERTY AMERICAN
    23. 25 TO 28 PRIS YD 0.5 ZD 0.5
    24. CONSTANTS
    25. MATERIAL CONCRETE ALL
    26. **********************
    27. ************************************************************
    28. SUPPORTS
    29. 1 TO 4 FIXED
    30. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    31. JOINT LOAD
    32. 17 FY -40
    33. 18 FY -20
    34. 19 FY -50 FZ 10
    35. 20 FY -30
    36. ***
```

```
37. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
38. JOINT LOAD
39. 17 FY -110 FZ -10
40. 18 FY -40
41. 19 FY -140 FZ 30
42. 20 FY -80
43. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
44. JOINT LOAD
45. 17 FX 20 FY 90
46. 18 FX 20 FY -90
47. 19 FX 30 FY 110
48. 20 FX 30 FY -120
49. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
50. JOINT LOAD
51. 17 FX -20 FY -90
52. 18 FX -20 FY 90
53. 19 FX -30 FY -110
54. 20 FX -30 FY 120
55. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
56. JOINT LOAD
57. 17 FY 30 FZ 30
58. 18 FY 30 FZ 20
59. 19 FY -40 FZ 10
60. 20 FY -30
61. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
62. JOINT LOAD
63. 17 FY -30 FZ -30
64. 18 FY -30 FZ -20
65. 19 FY 40 FZ -10
66. 20 FY 30
67. ********************************************************************************
68. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
69. ************************************************************
70. LOAD COMB 201 1.0DL+1.0LL
71. 1 1.0 2 1.0
72. LOAD COMB 202 1.0DL+0.75LL
73. 1 1.0 2 0.75
74. LOAD COMB 203 1.0DL+1.0WLX1
75. 1 1.0 3 1.0
76. LOAD COMB 204 1.0DL+1.0WLX2
77. 1 1.0 4 1.0
78. LOAD COMB 205 1.0DL+1.0WLZ1
79. 1 1.0 5 1.0
80. LOAD COMB 206 1.0DL+1.0WLZ2
81. 1 1.0 6 1.0
82. LOAD COMB 207 1.0DL+0.75LL+0.75WLX1
83. 1 1.0 2 0.75 3 0.75
84. LOAD COMB 208 1.0DL+0.75LL+0.75WLX2
85. 1 1.0 2 0.75 4 0.75
86. LOAD COMB 209 1.0DL+0.75LL+0.75WLZ1
87. 1 1.0 2 0.75 5 0.75
88. LOAD COMB 210 1.0DL+0.75LL+0.75WLZ2
89. 1 1.0 2 0.75 6 0.75
90. LOAD COMB 211 0.6DL+1.0WLX1
91. 1 0.6 3 1.0
92. LOAD COMB 212 0.6DL+1.0WLX2
```

```
 93. 1 0.6 4 1.0
 94. LOAD COMB 213 0.6DL+1.0WLZ1
 95. 1 0.6 5 1.0
 96. LOAD COMB 214 0.6DL+1.0WLZ2
 97. 1 0.6 6 1.0
 98. ********
 99. ********
100. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**8**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 4/ 4/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112058.6 MB

```
101. FINISH

          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= APR  8,2020   TIME= 11:43:21 ****
```

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
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```